MECHANICAL PROPERTIES OF MATERIALS: Fundamentals, elastic and plastic behaviour

Mechanical properties of materials

- Materials choice
- Design issues
- Correlation requested properties/class of material

- Mechanical **properties independent of time**
- Mechanical **properties dependent on time**
- Mechanical **properties dependent on temperature**

σ **(STRESS)** and ε **(STRAIN) : definition**

σ **(STRESS) = F** (force, Newton, N) / **A** (surface, m^2)
σ **= F/A** **(**N/m^2 = Pa) (**N/mm^2 = MPa**)

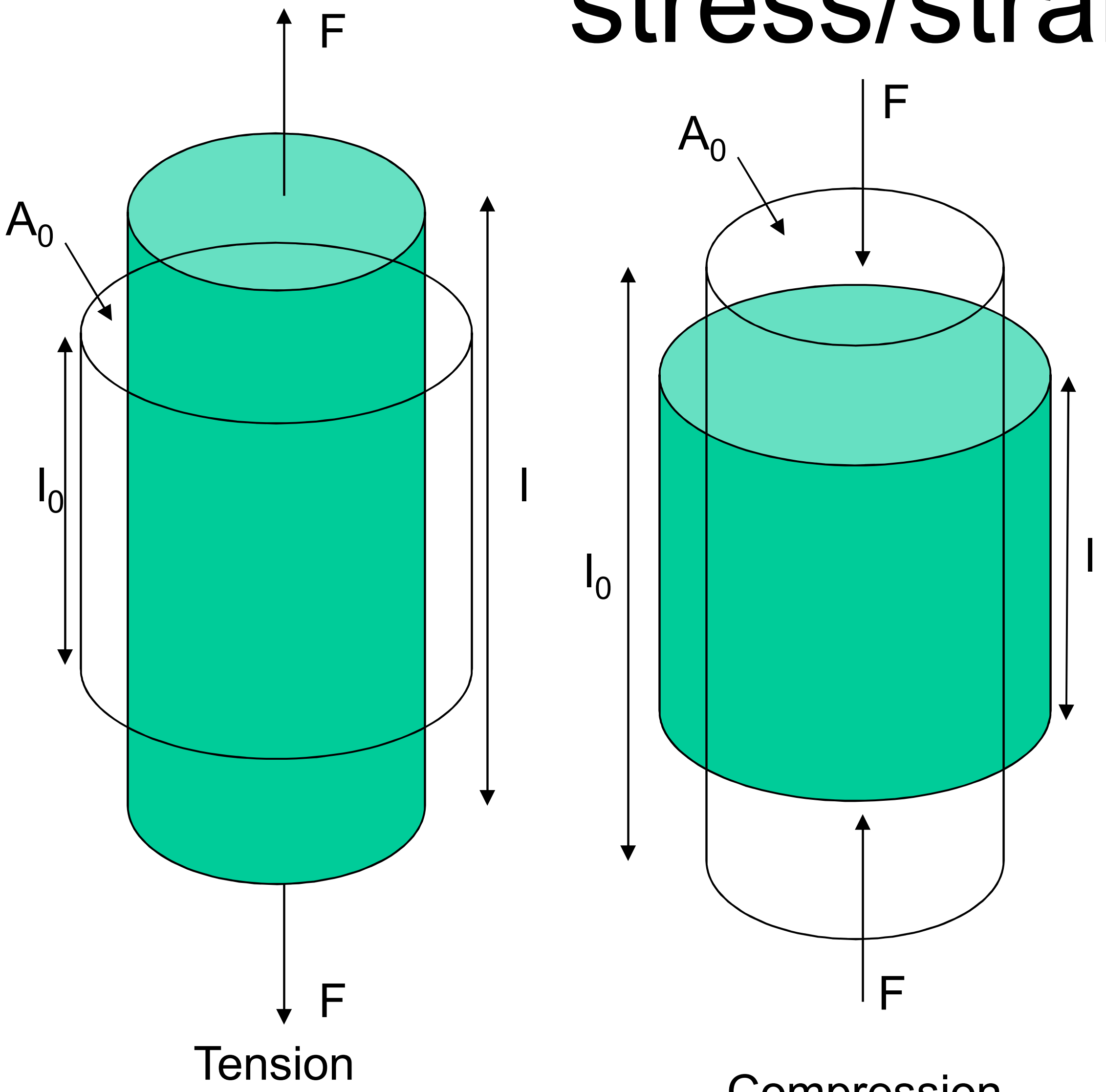
DEFORMATION or STRAIN:
(dimensionless or %)

$$\varepsilon = (l - l_0)/l_0 = \Delta l/l_0$$

sample length = l
initial sample length = l_0

tensile stress, compressive stress, bending stress, shear stress

Tensile and compression stress/strain

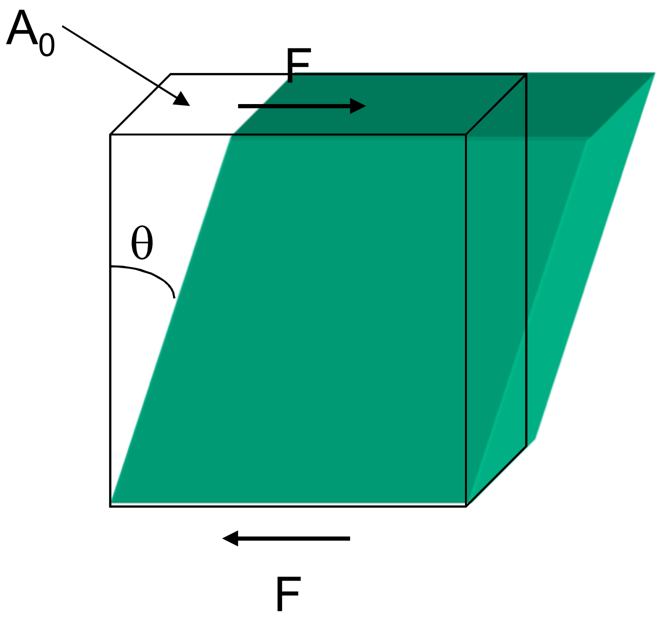


Stress, $\sigma = F/A_0$

Strain, $\varepsilon = (l - l_0)/l_0$

$= \Delta l/l_0$

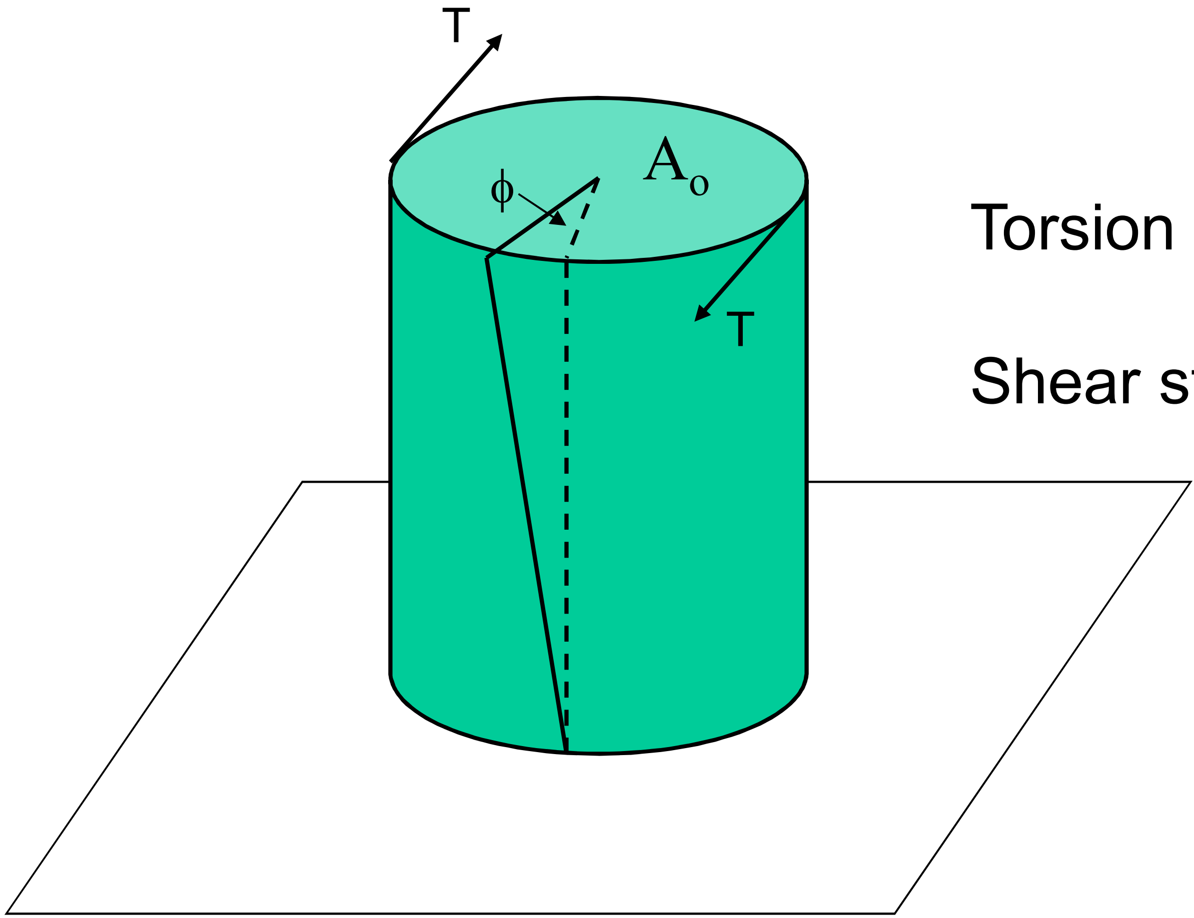
Shear



(Shear stress), $\tau = F/A_0$

(Shear strain), $\gamma = \tan\theta$

Torsion

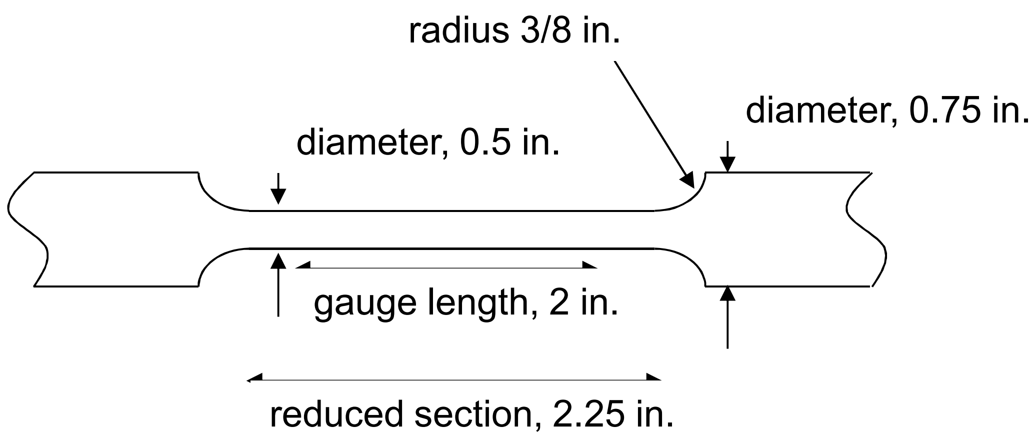


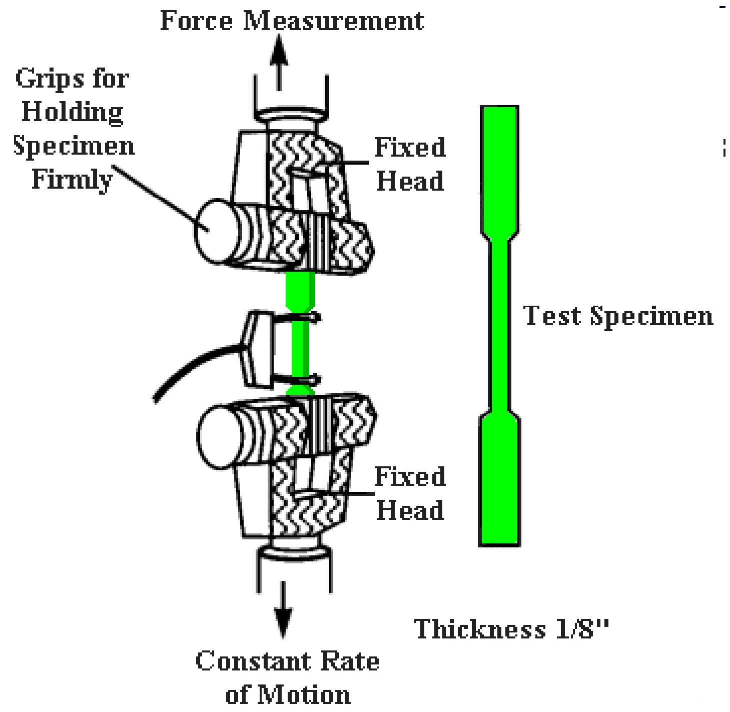
Torsion $\tau = T/A_0$

Shear strain, $\gamma = \tan\phi$

Stress-strain curves (tensile test)

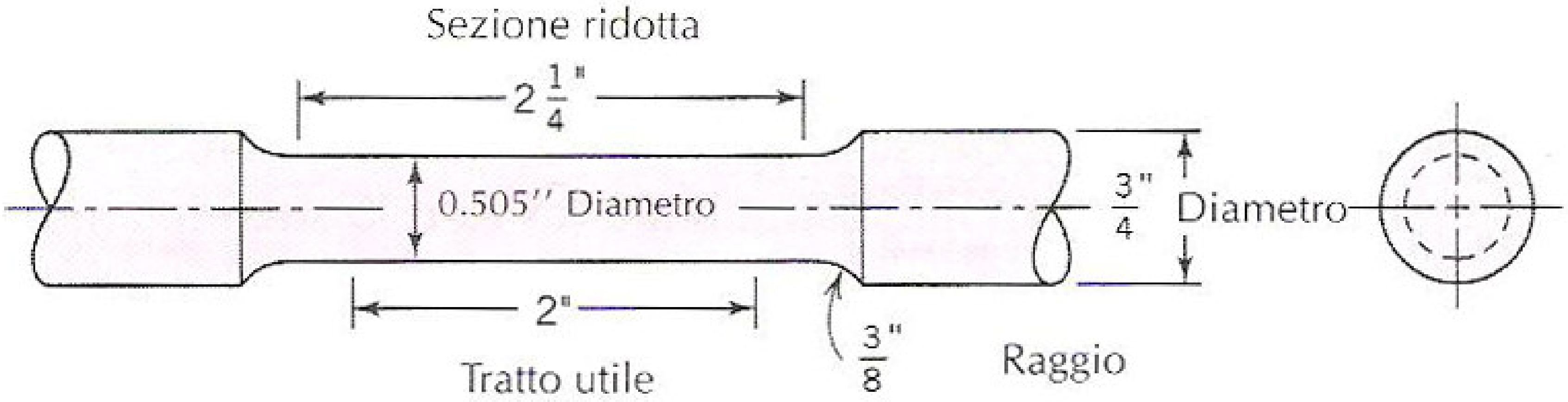
- Standard tests!

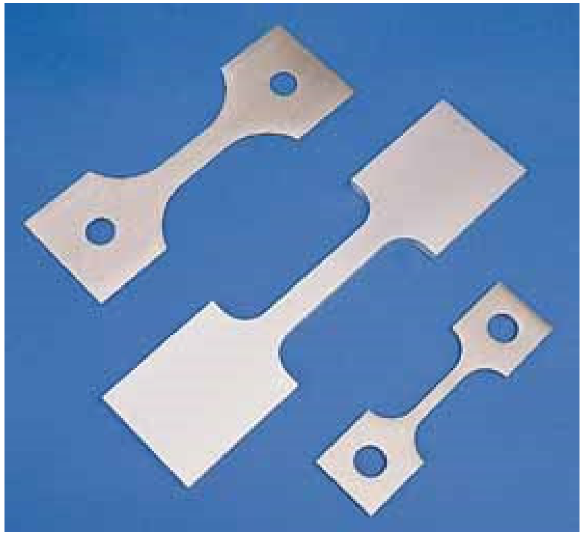




ASTM (American Society for Testing and Materials)
ISO
UNI EN

Standard circular cross-section for tensile tests





Metallic specimens

For most of materials, low stresses correspond to proportionally low deformations (direct proportionality)

$\sigma = \mathbf{E}\,\varepsilon$ **(Hooke law, E=Young modulus or elastic modulus)**

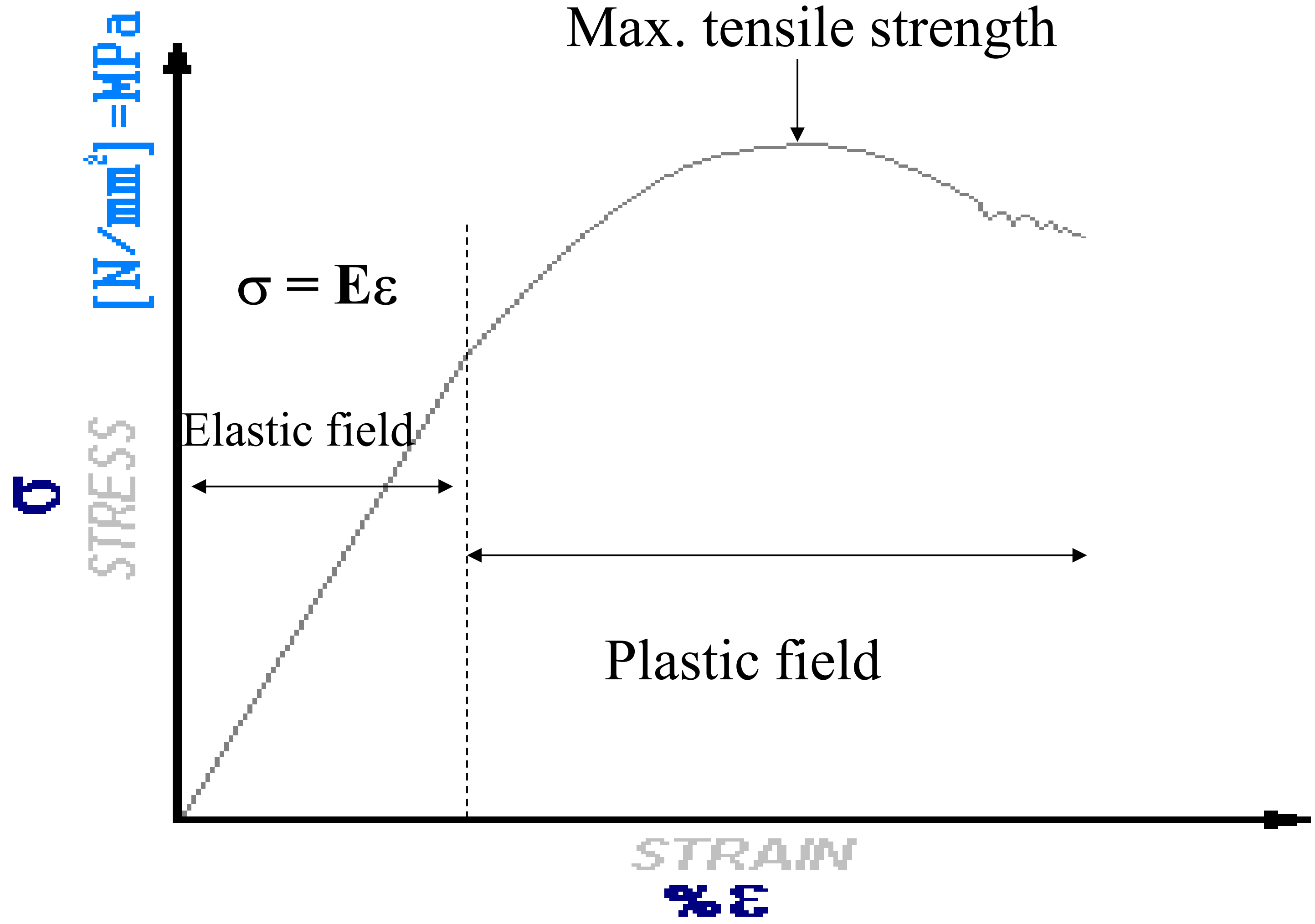
$\sigma = F/A$

$\varepsilon = \Delta l/l_o$

$[E] = \text{MPa or GPa}$

$F/A = E\,\Delta l/l_o$

Tensile test curve on a metal



Elastic behaviour

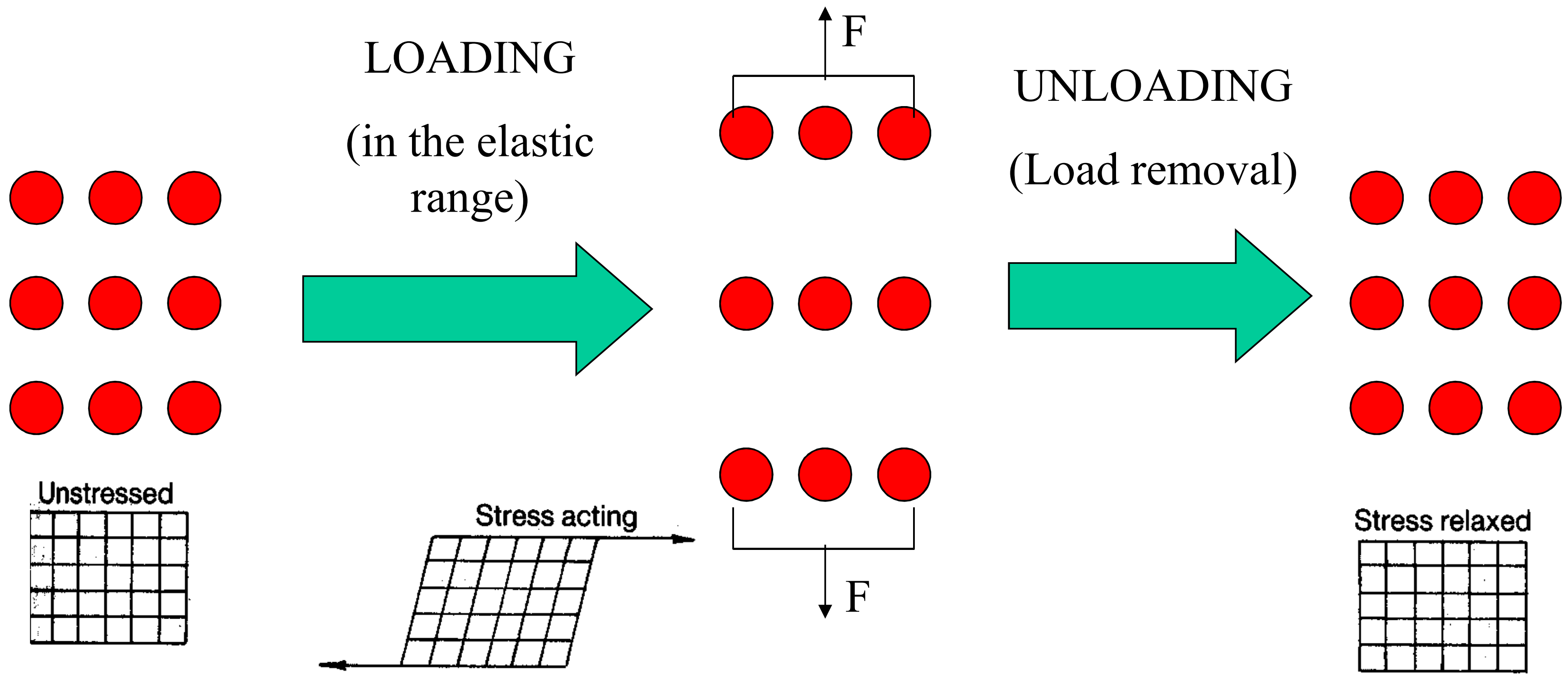
- Small stress corresponds to small deformation having a direct proportionality between them.
- Small deformations are immediately recovered when stress is removed
- **$E = \sigma / \varepsilon$ = [N/mm^2] = [MPa]**

Young's modulus or Elastic Modulus **E**

(Hooke law)

Elastic modulus and atomic bond

- $E_{ceramics} > E_{metals} >> E_{polymers}$
- Elastic modulus is directly proportional to the energy of the atomic bond (and melting temperature)
- No fracture of bonds in the elastic field
- Increasing of the bond length when stress is applied.
- Immediate recovering of the bond equilibrium length after stress removal



EQUILIBRIUM STATE

NO APPLIED LOAD

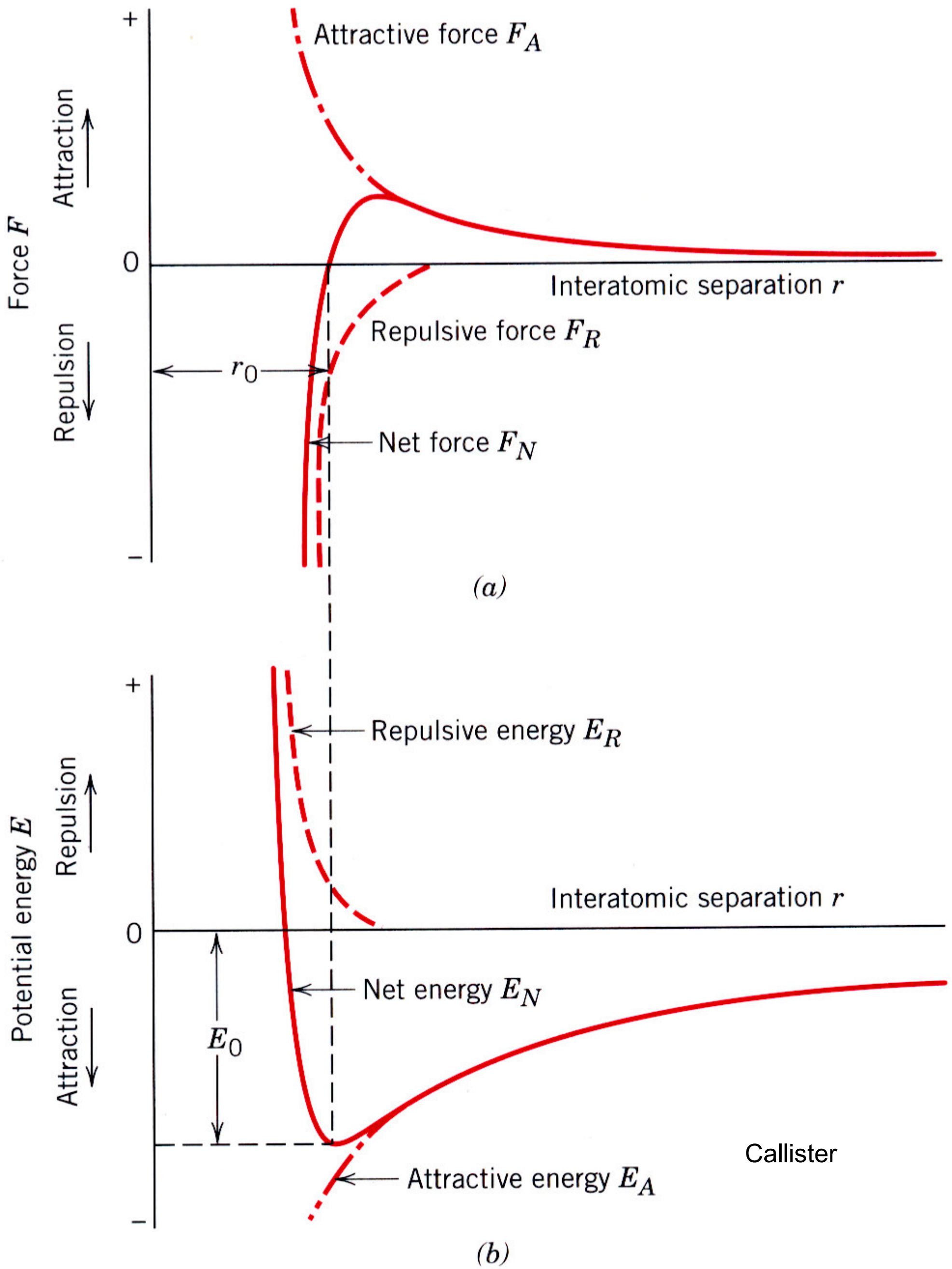
THE DISTANCE BETWEEN ATOMS IS THAT OF THE **STABLE CHEMICAL BOND**

DURING ELASTIC DEFORMATION THE BONDS BECOME **LONGER** IN THE DIRECTION OF LOAD APPLICATION

(much longer as weaker the bonds are)

WHEN THE LOAD IS REMOVED THE **ATOMS COME BACK TO THEIR EQUILIBRIUM DISTANCE**

Condon-Morse curves

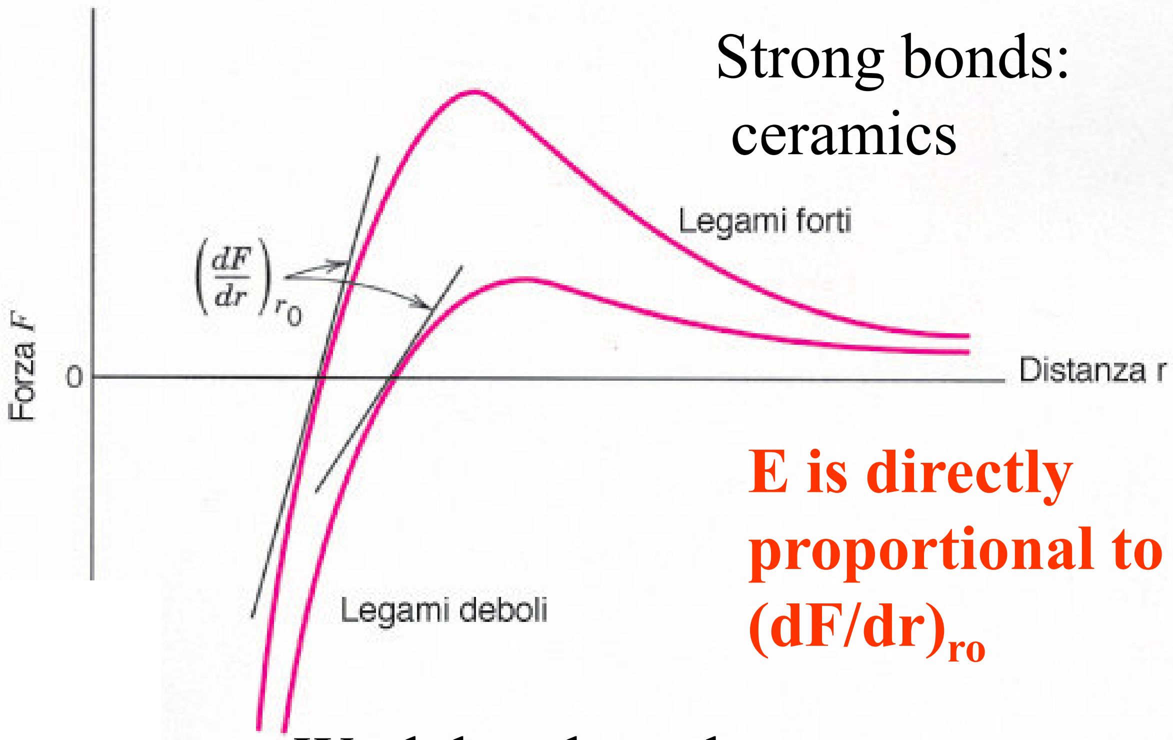


Callister

How atomic bonds are responsible of materials properties

Elastic modulus and bond strength

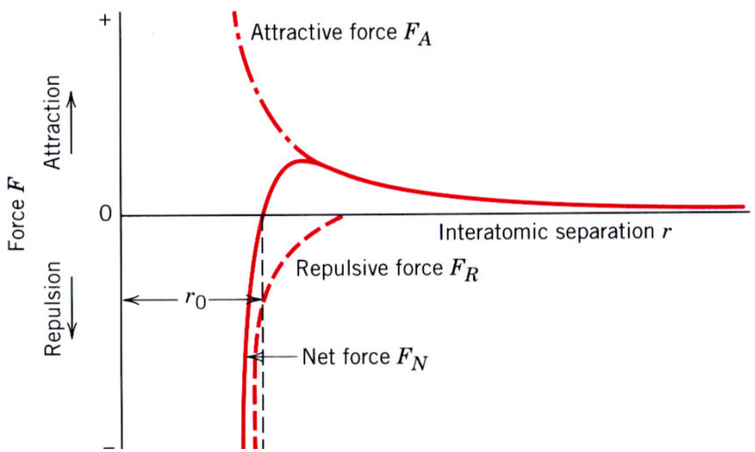
FIGURA 6.7
Curva forza–distanza interatomica per materiali con legami deboli e forti. Il valore del modulo elastico è proporzionale alla pendenza di ciascuna curva calcolata alla distanza interatomica di equilibrio r_0.



Strong bonds: ceramics

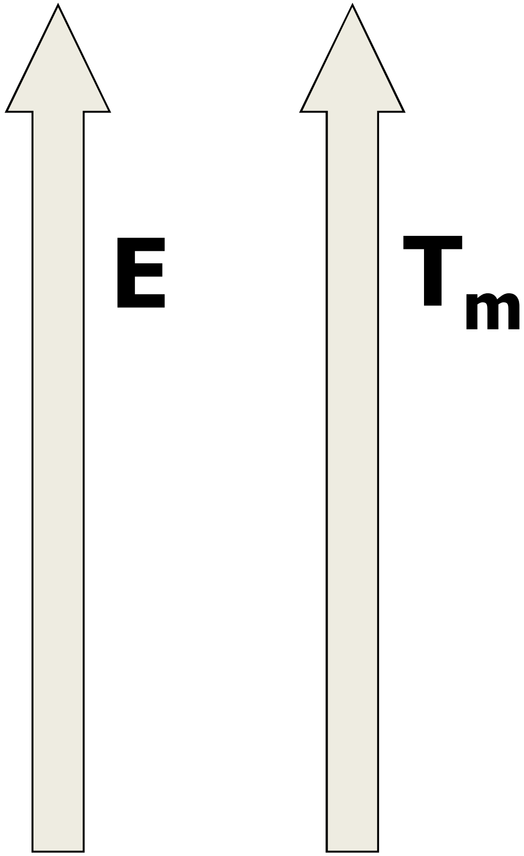
Weak bonds: polymers

E is directly proportional to $(dF/dr)_{ro}$



a introduzione

Elastic modulus and melting temperature

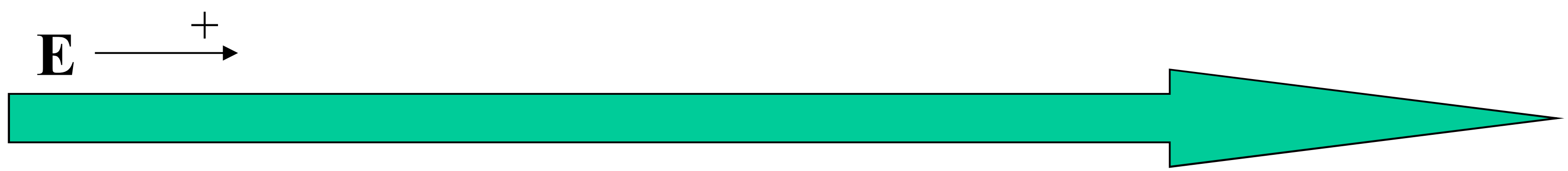


Material	Temp melting [° C]	E (GPa)
TiC	3160	310
Al_2O_3	2045	370
W	3410	393
Fe	1536	207
Cu	1083	111
Al	660	70
Pb	327	14
Polyetilene	130	1
glasses	-	70-90

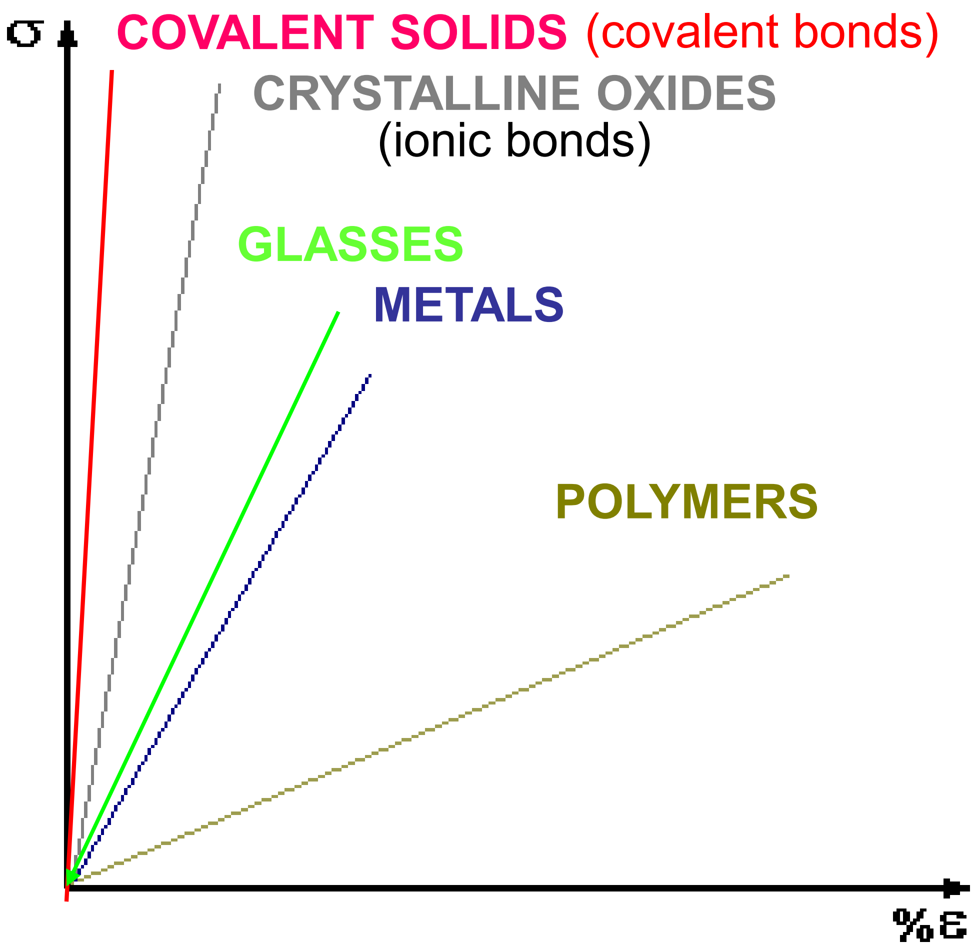
Materiale	E (GPa)	Materiale	E (GPa)
Diamond	10^3	**Ti and its alloy**	85-130
WC	500-600	**Zn and its alloy**	45-95
SiC	450	**Al and its alloy**	70-80
Al_2O_3	300-400	**Mg and its alloy**	40-45
TiC	320	**GFRP ***	10-40
Ni	215	**Wood $\parallel$ to fibers**	10-15
Steel	195-215	**Wood $\perp$ to fibers**	0,5-1
CFRP**	100-200	**Plastics**	10^{-1}-5
Copper and its alloy	120-150	**Expanded plastic**	10^{-3}-10^{-2}

***Glass fiber reinforced polymers**

****C fiber reinforced polymer**



Polymers Metals Ceramics



Young's modulus depends on some parameters:

- Bonding energy
Stronger chemical bonds
→ higher bonding energy
→ higher **E**

- Amorphous vs. crystalline structure
Amorphous materials show lower **E** than crystalline ones

More than 4 orders of magnitude between **E** of diamond and **E** of polymers!

Which of these materials is «BRITTLE»?
Which is «FRAGILE»?
Why?

The elastic modulus E is assumed to be (almost) independent of alloying elements (e.g. interstitial atoms) → E = 210 GPa

Low amount of alloying elements → the mean energy of the bond remains constant → E = cost

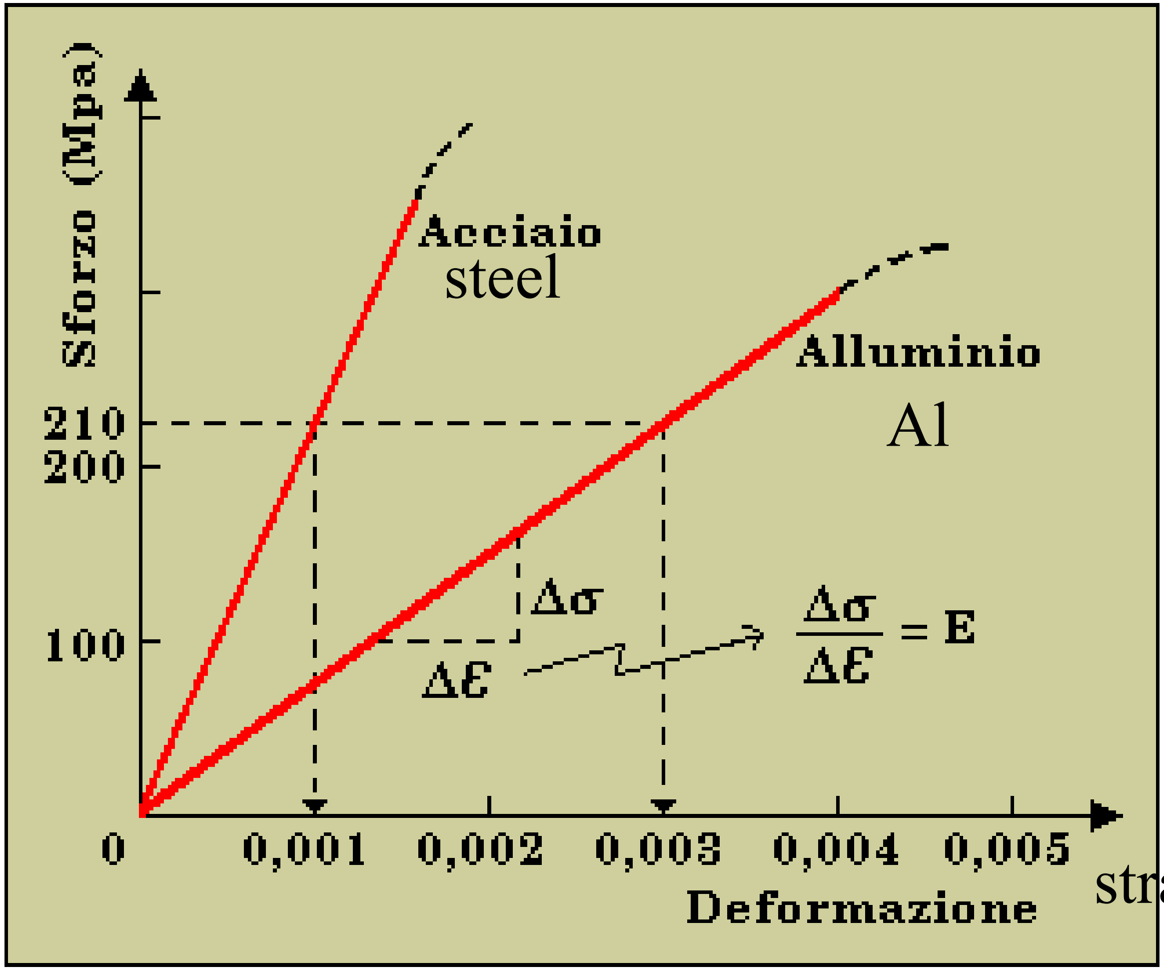
High amount of alloying elements → E = 190-230 GPa

THE ELASTIC MODULUS IS NOT RELATED TO THE MOVEMENT OF DISLOCATIONS!

		Modulo di elasticità alla trazione E
		N/mm^2
Ferro	Fe 37/360	190000
Ferro	Fe 430	200000
Ferro	Fe 510	210000
Acciaio non legato	C40	220000
Acciaio non legato	C 45	220000
Acciaio legato	18NiCrMo5	230000
Acciaio legato	34CrNiMo6	220000
Acciaio legato	42 CrMo 4	230000
Acciaio per cilindri	St35 - St37	200000
Acciaio per cilindri	ST 52	220000
Acciaio per cilindri	ST E 460	220000
Acciaio INOX	AISI 304 X5CrNi 18-10	196000
Acciaio INOX	AISI 316 X5CrNiMo 17-12-2	196000
Acciaio INOX	AISI 410 X12 Cr 13	198000
Acciaio INOX	AISI 420 X30 Cr 13	198000
Acciaio INOX	AISI 430 X6 Cr 17	200000
Acciaio INOX	AISI 630 X5CrNiCuNb 16-4	196000

How to measure the elastic modulus: from tensile curve

stress



strain

How to measure the elastic modulus

– Drawbacks of tensile tests:

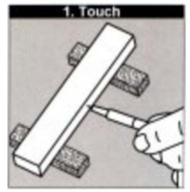
❖Sample preparation

❖Destructive test

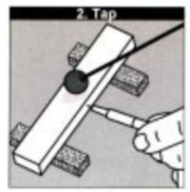
❖Unsuitable for brittle materials

- **Measurement of the elastic modulus by compression or bending tests, indentations…**
- **Non destructive tests**

TOUCH



TAP

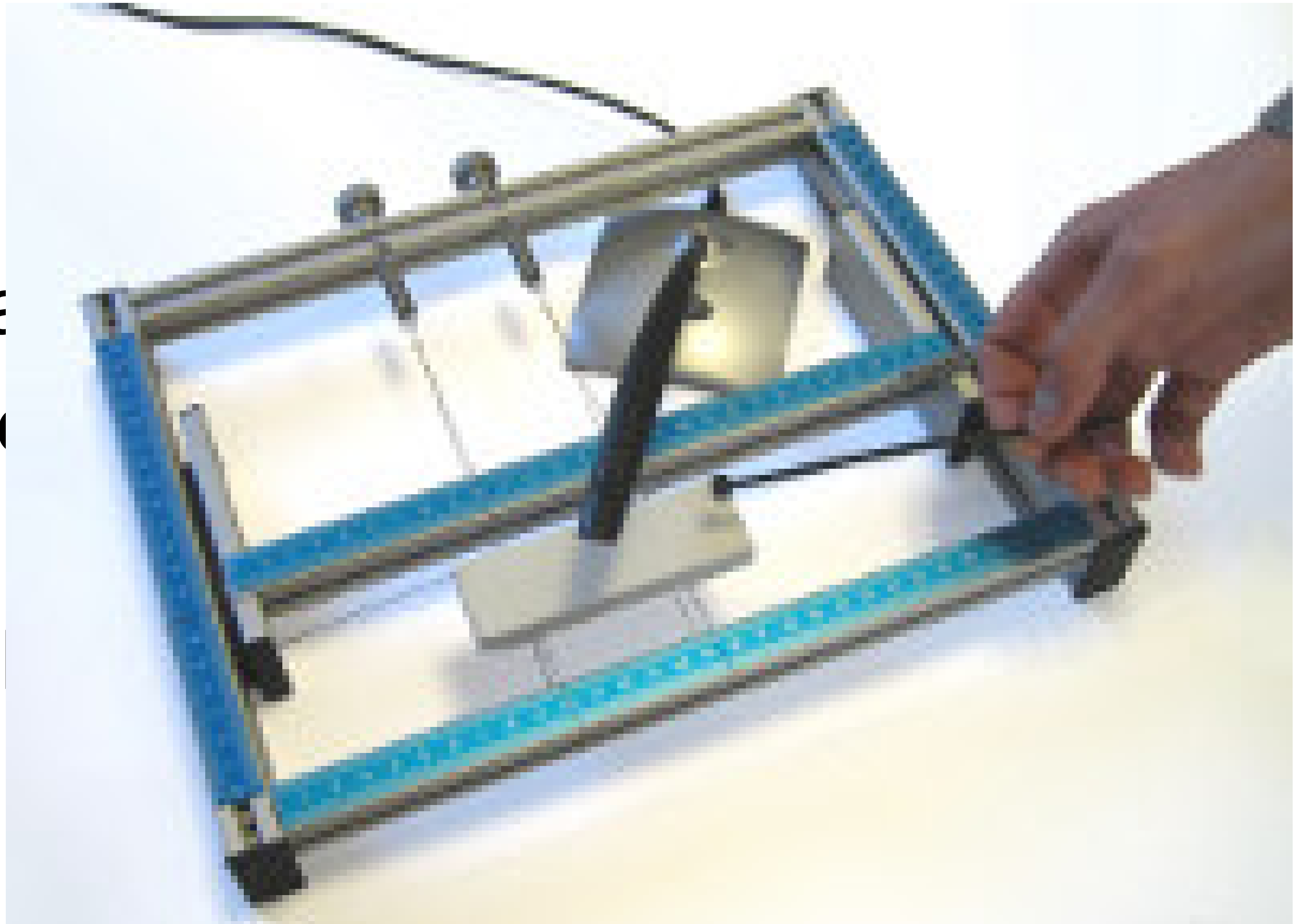


READ

Elastic modulus by sonic test

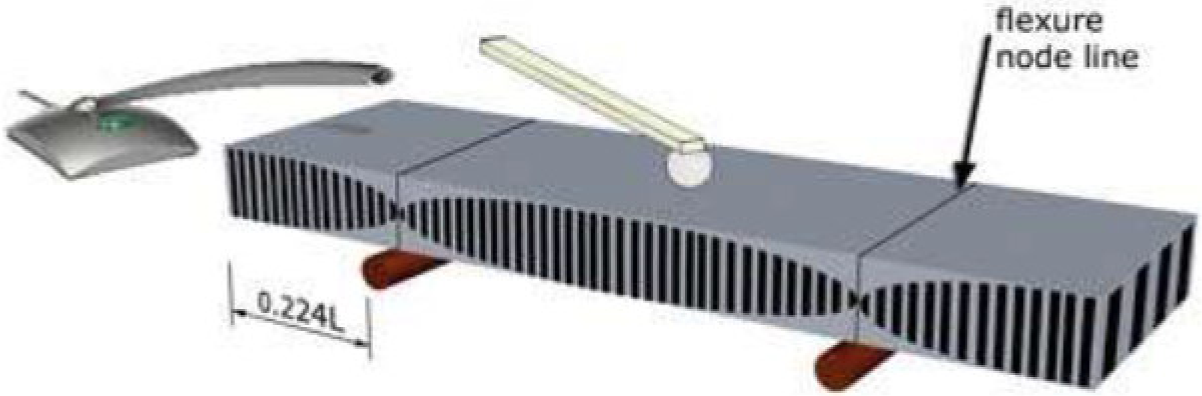
- **E= 0.9465 * ($\Gamma \nu^2 L^4 \rho / t^2$)**

- Γ = costant
- ν = sonic wa
- L= sample l
- ρ = material
- t = sample t



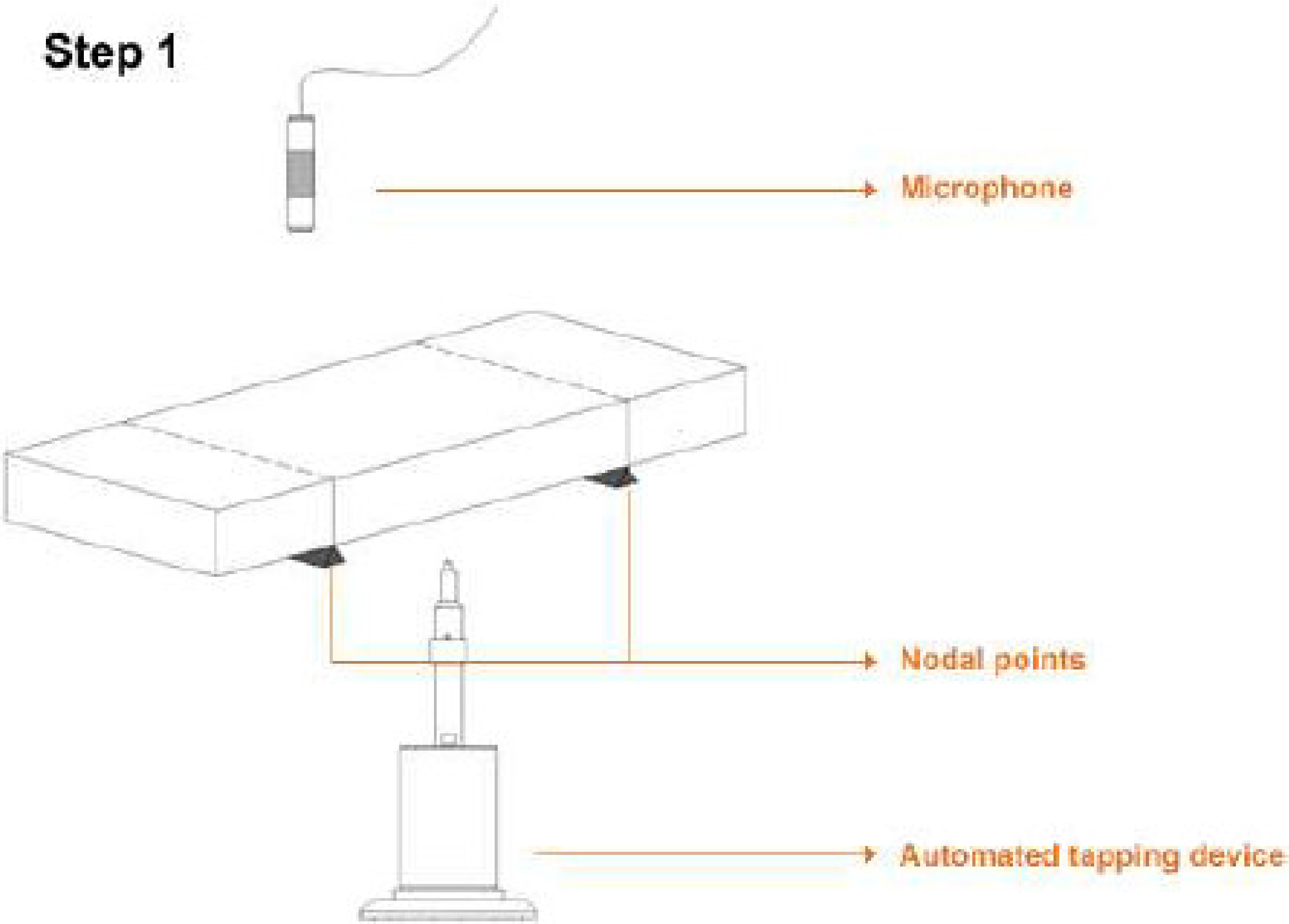
For photos and technical specifications pleas visit: www.imce.net/presentation

The Principle of Impulse Excitation



- Figure 1. A method for supporting a rectangular bar to induce the flexural mode of vibration. The sample is supported on two knife edges placed at the standing wave nodes, which occur at 0.224 of the length from each end. The bar then flexes freely in and out of its plane, as indicated by the vertical lines.
- ASTM standards E1867 and C1259

Step 1



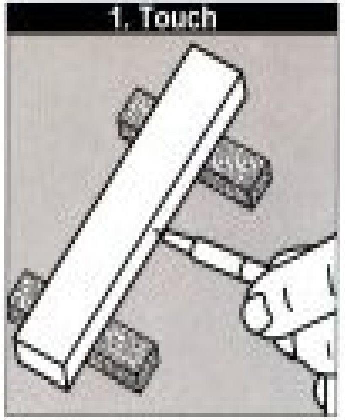
Measuring elastic modulus (E) by GrindoSonic

☺ Useful for highly porous samples

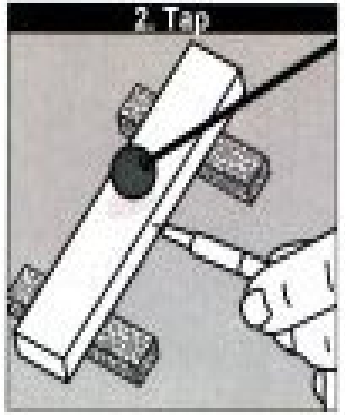
☺ Very useful for quality control!

☺ No limtiations on size

☹ Limitations on geometry



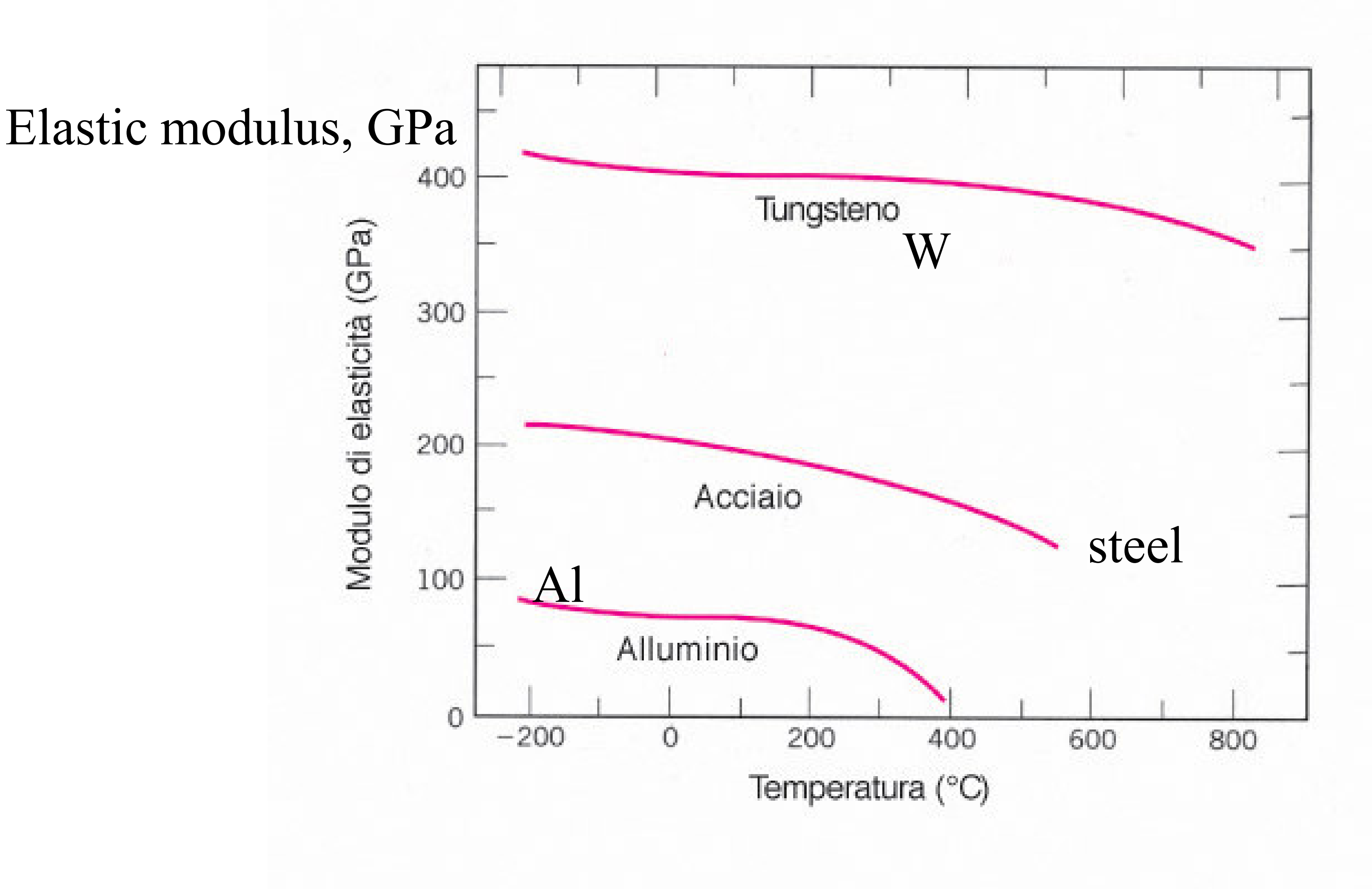
Touch
In the majority of cases, a piezo-electric detector is used to capture the vibrations, and to convert it into an electrical signal. The point of the detector is simply brought into contact with the sample.



Tap
The sample is excited into vibration through the means of a light tap. Very little energy is required.

Read
Almost immediately the numerical result is displayed on the front panel. A few moments later the display is cleared and the instrument is ready for the next measurement.

Elastic modulus vs T



ELASTIC MODULUS IN COMPOSITES

Mixture rule: elastic modulus is a weighted average, based on the volumetric fraction, of the elastic moduli of the components:

$$E_c = E_1 \cdot f_{v1} + E_2 \cdot f_{v2} + \ldots E_n \cdot f_{vn}$$

Composite with **50% vol. C fibres + 50% vol. polymeric matrix**
Carbon fibers: E ≈ 200 GPa
Polymer: E ≈ 0.6 GPa
Composite **E ≈ (200*0.5+0.6*0.5) ≈ 100 GPa**

Composite **30% vol. C fibres + 70% vol. polymeric matrix**
Composite **E≈ (200*0.3 + 0.6*0.7) ≈ 60 GPa**

Elastic modulus of a steel vs T

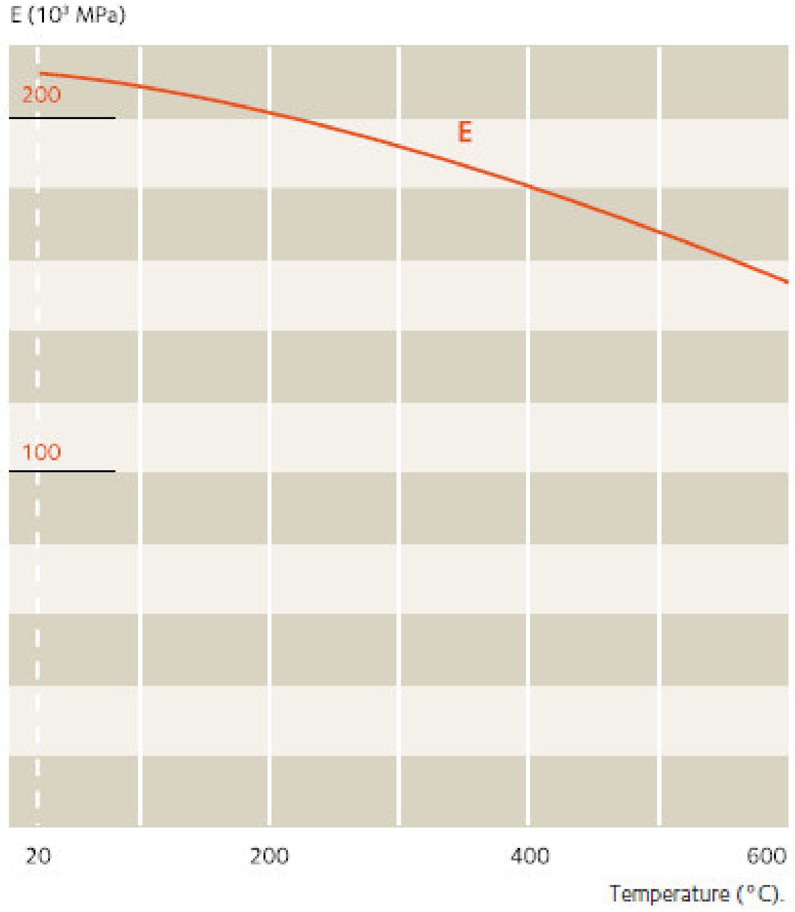
Physical properties

(Cold rolled sheet - annealed*)

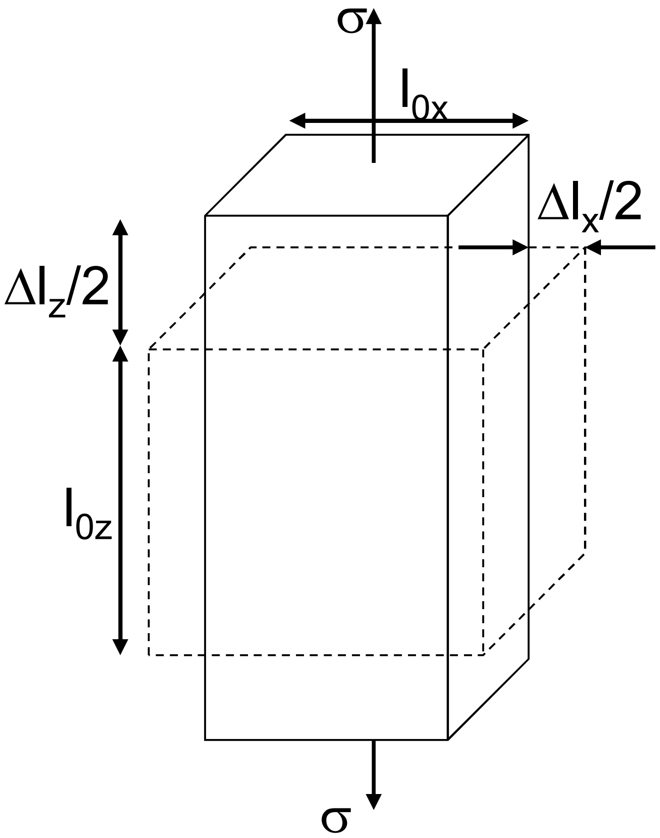
Density	d	kg/dm³	20°C	7.7
Melting temperature		°C		1500
Specific heat	c	J/kg.K	20°C	450
Thermal conductivity	k	W/m.K	20°C	21.3
Mean coefficient of thermal expansion	a	10^{-6}/K	20-200°C 20-400°C 20-600°C 20-800°C	11.5 12 12.6 13.5
Electric resistivity	p	Ω.mm²/m	20°C	0.70
Magnetic permeability	μ	at 0.8 kA/m DC or AC	20°C	550
Young's modulus	E	MPa.10³	20°C	210

* Typical values.

Young's modulus at high temperature



Poisson coefficient (ν)



$$\upsilon = -\frac{\varepsilon_x}{\varepsilon_z}$$

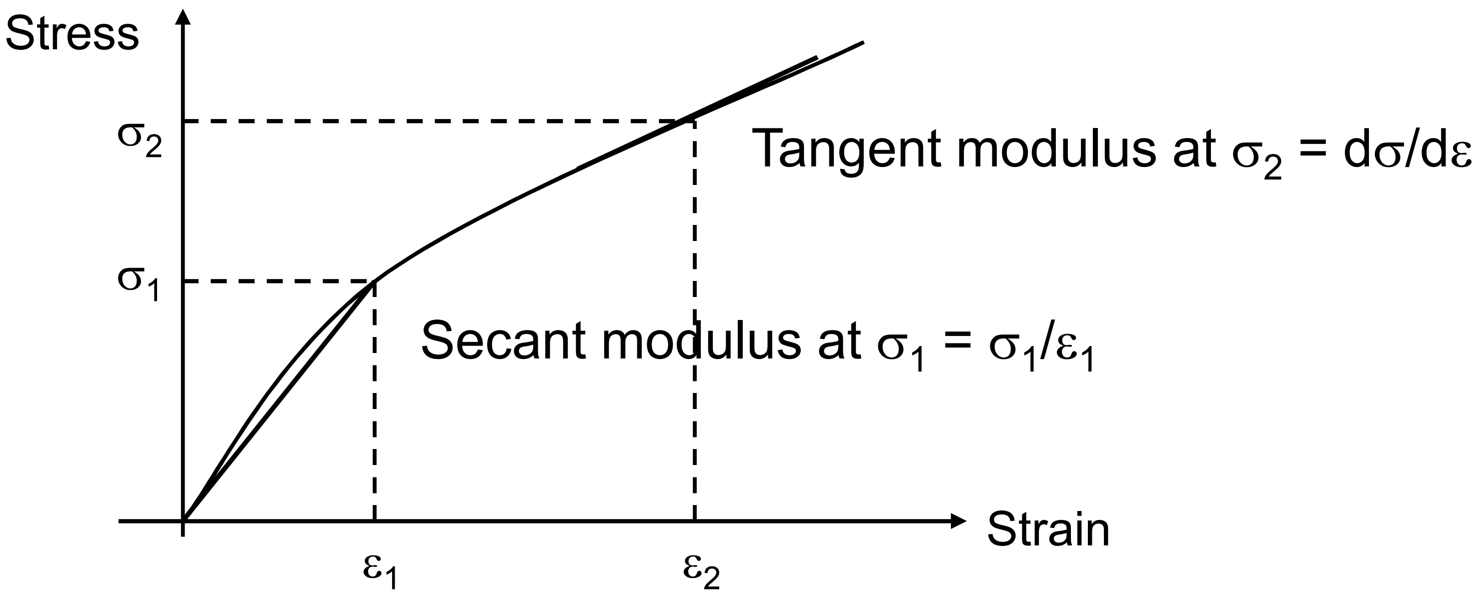
$$= -\frac{\varepsilon_y}{\varepsilon_z}$$

Metals: about 0.33
Ceramics: 0.17 - 0.27
Polymers: 0.33 - 0.5
The Poisson coeff. is always < 0.5

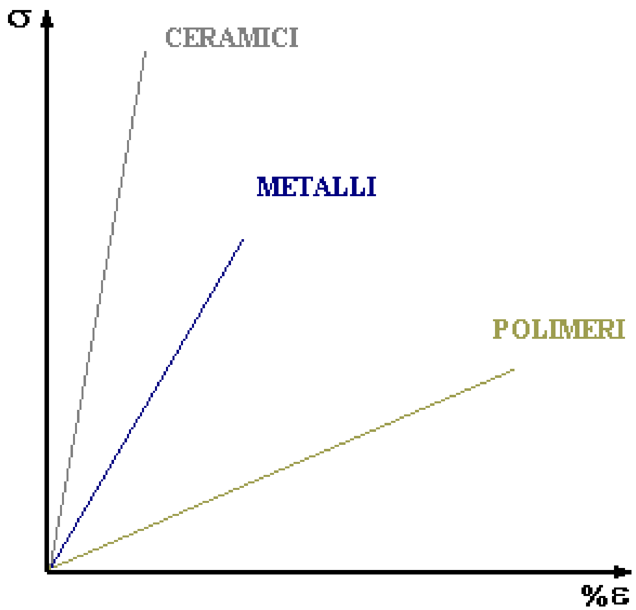
- ε_x, ε_y and ε_z have opposite sign

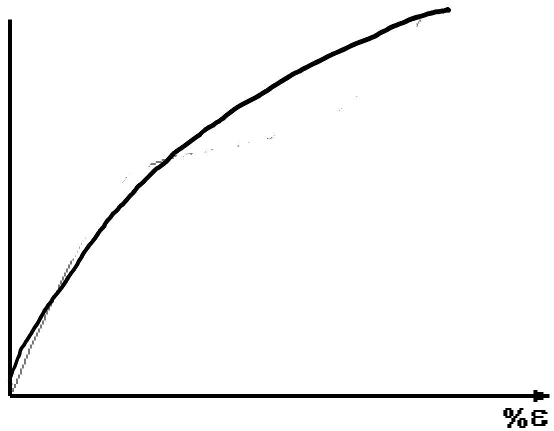
Non linear elastic behaviour

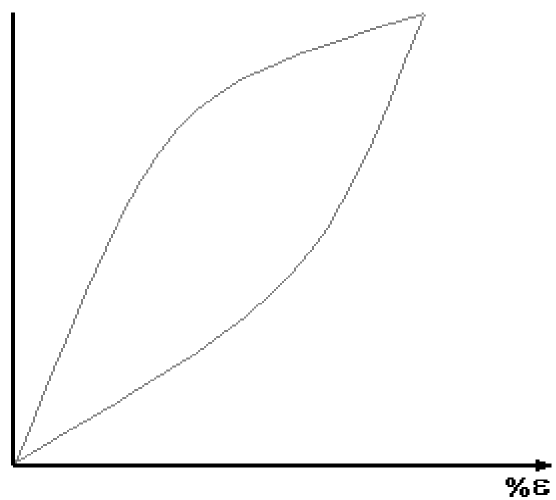
- Some polymers, cast iron, concrete,….
- E calculated locally



Summary







Linear elastic Behaviour
(most materials)

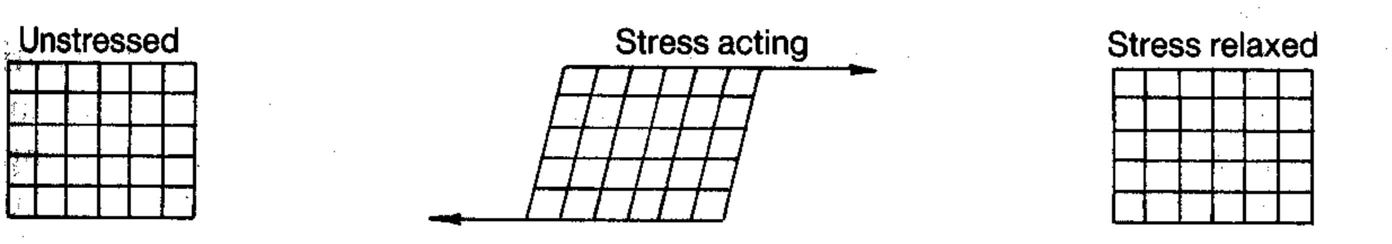
Non linear elastic behaviour
(some polymers, cast iron, concrete,…)

Anelastic behaviour
Time is required to recover deformation after stress removal
(some polymers)

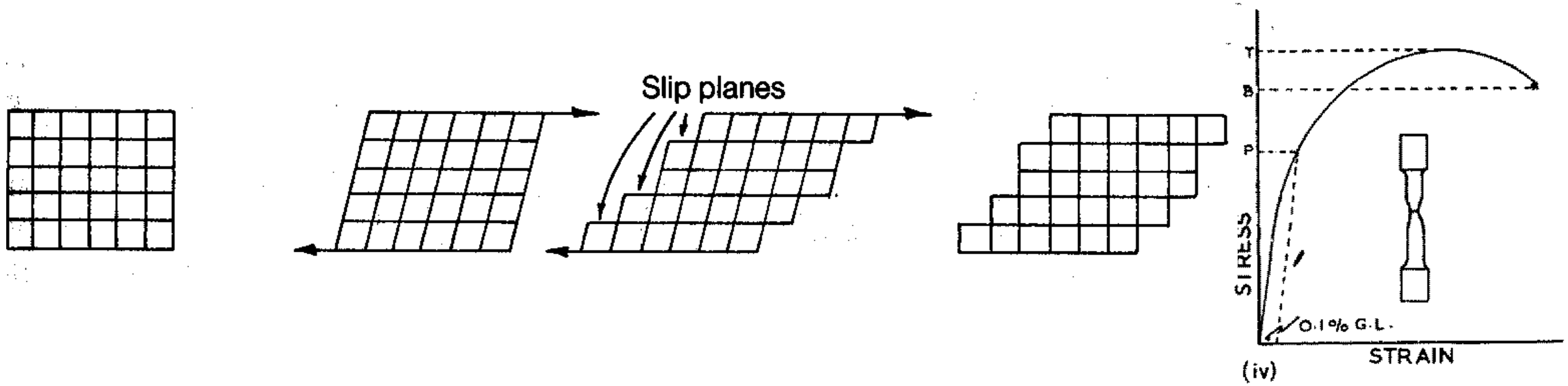
Plastic deformation and chemical bond

- When applied stresses increases and corresponding deformations are higher than about 0.002, **the Hooke law does not work any more.**
- Bonds start breaking and, **if possible**, other bonds are formed in different positions
- If the load is removed, the sample recovers only the elastic deformation, but **a plastic, permanent deformation is evident.**
- The frontier between elastic and plastic deformation is defined **yield strength or limit of proportionality**

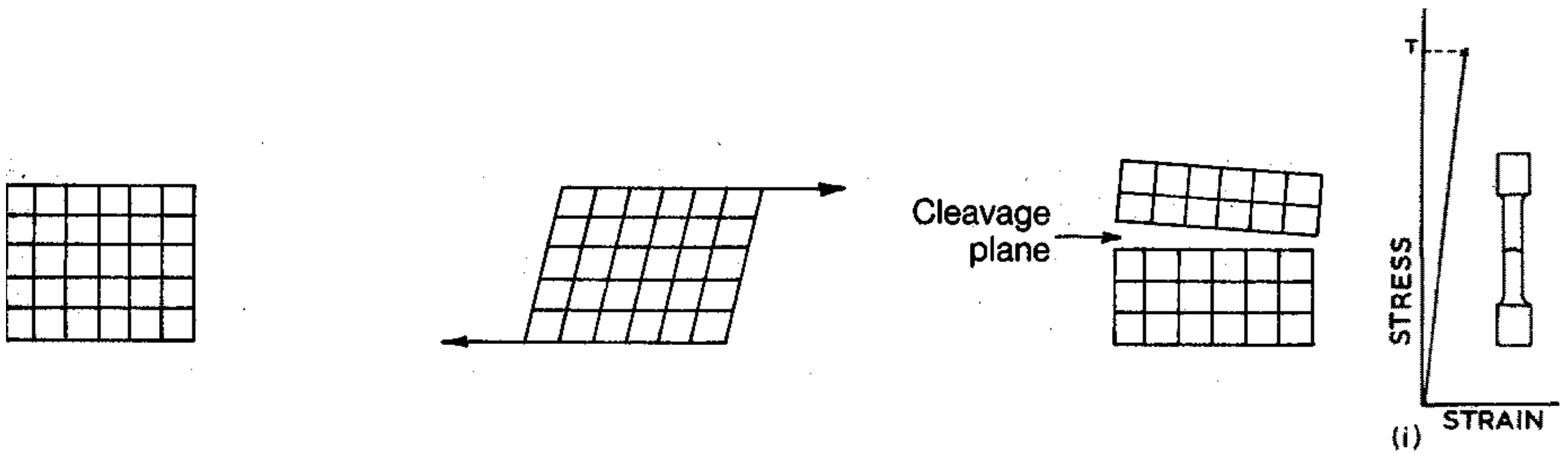
ELASTIC DEFORMATION



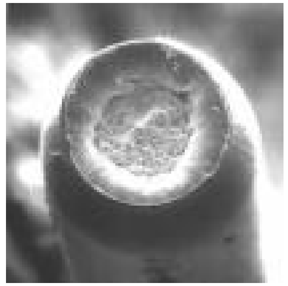
PLASTIC DEFORMATION

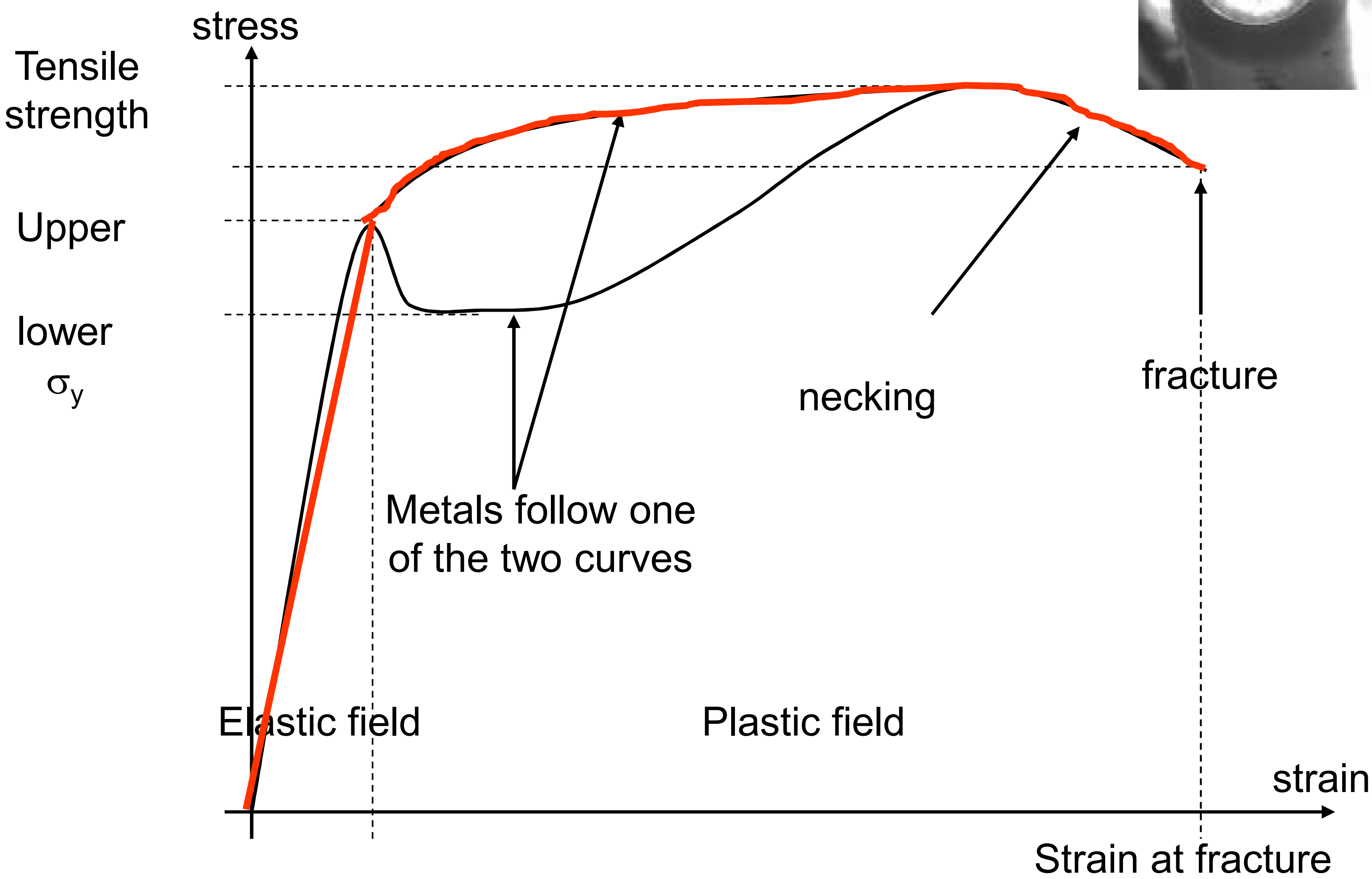


BRITTLE FRACTURE (CLEAVAGE)



σ–ε curve for metals

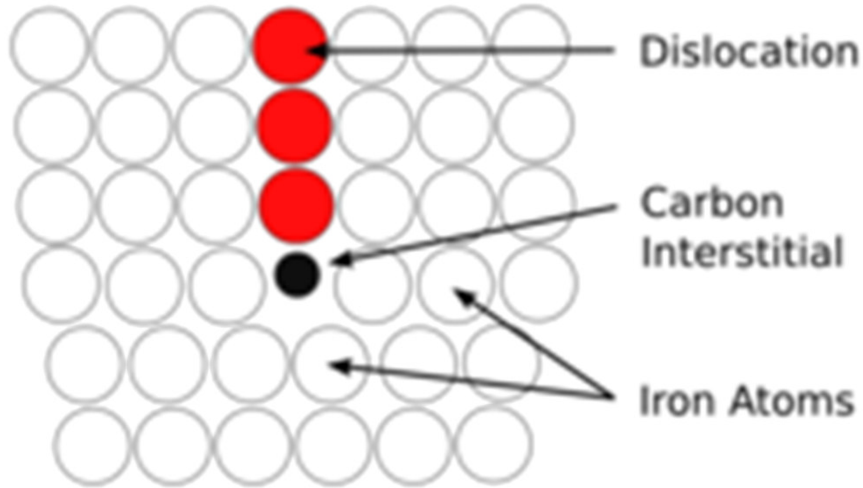


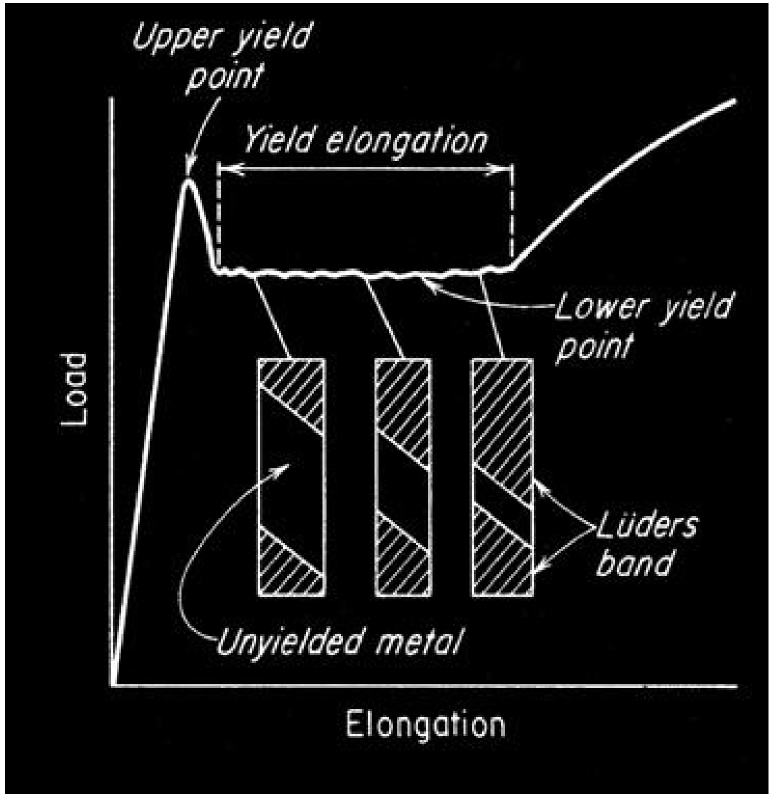


In steel we have interstitial C, causing the dislocations to pin (to stop).

Thus, it becomes difficult for dislocation to move and we need higher load to make them to move (upper yield point).

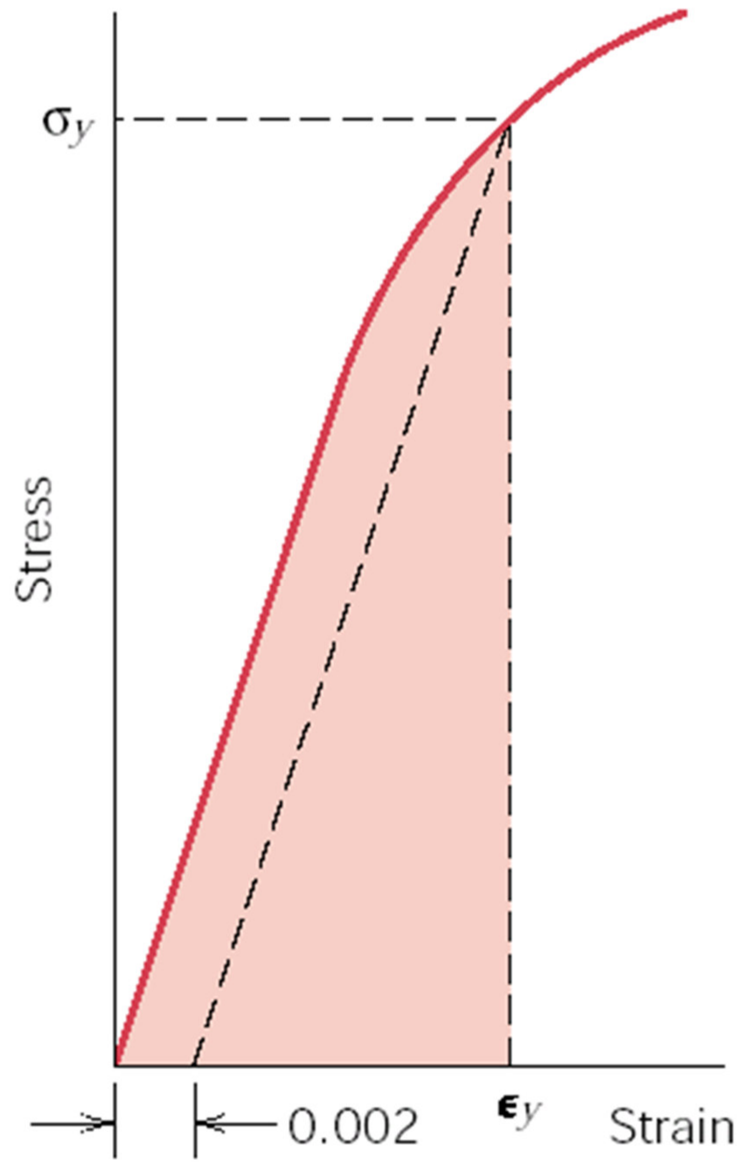
Then, the pinning will be gone and dislocations can move freely and we need smaller load to keep them moving (plastic deformation going) → lower yield point.





Yield strength at 0.2 % strain (0.002)

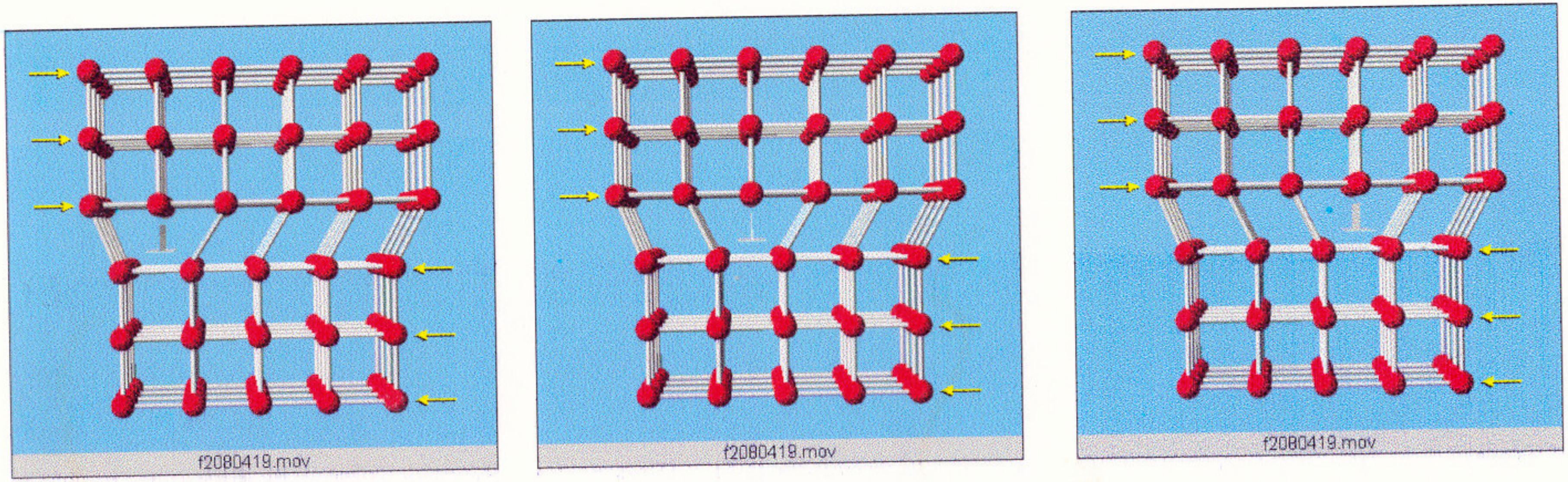
This type of monotonic curve is observed, for example, in pure metals, FCC alloys, Al alloys.

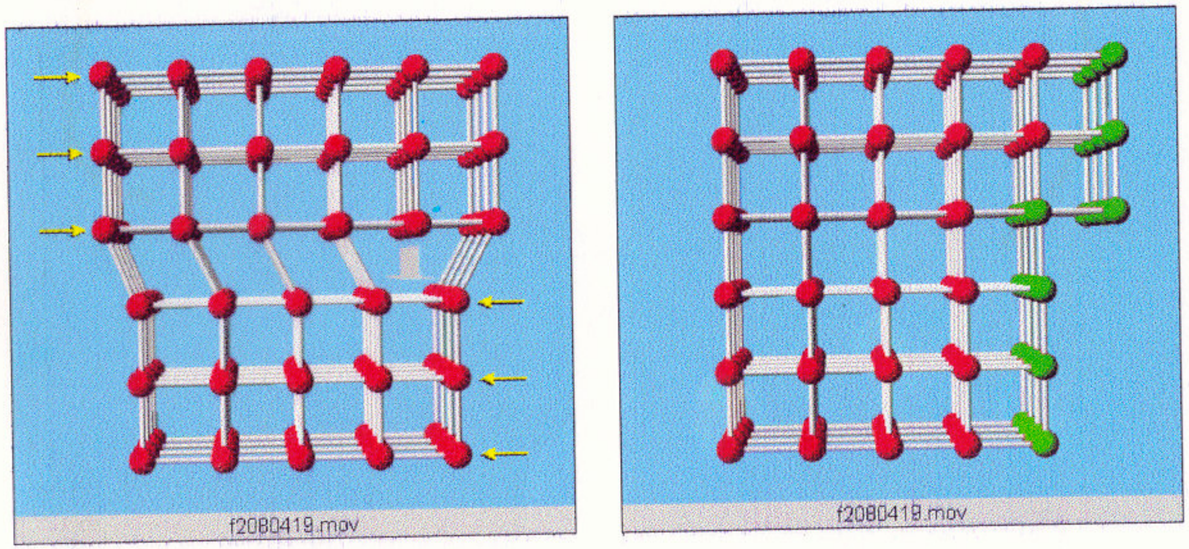


Yield strength for different materials

	σ_y [MPa]
steels	250-1400
Ni alloys	200-1600
Ti alloys	200-1300
Cu alloys	60-950
Al alloys	100-650
Mg alloys	80-300
Epoxy resins	30-100
nylon	50-90
polystirene	35-70
ABS	50
polyethylene	6-30

Yield strength and dislocations





Dislocations explain the 3 orders of magnitute between **theorethical and experimental yield strength**

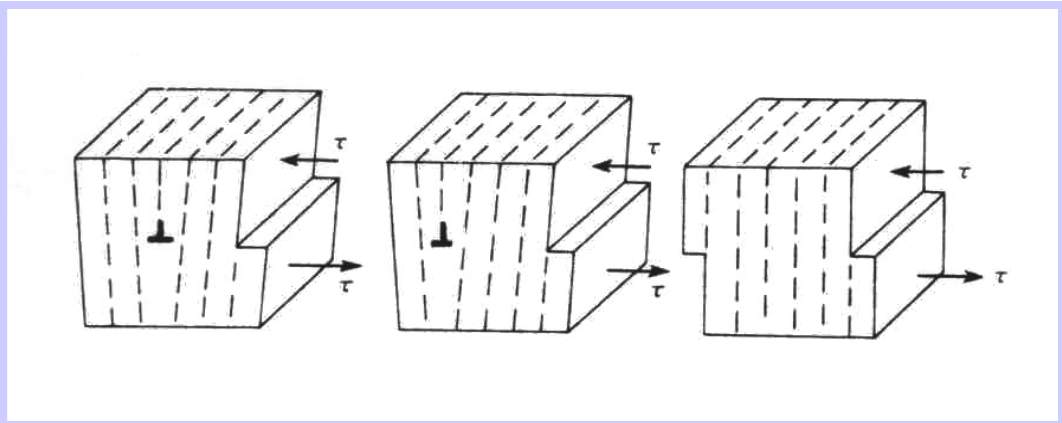
Without dislocations, the yield strength would be about **E/8 (GPa)**

Material	Tensile Yield Stress (psi)
pure (99.45%) annealed Al	4×10^3
pure (99.45%) cold drawn Al	24×10^3
Al alloy - precipitated, hardened	50×10^3
a "perfect" single crystal of pure Al	ca. 10^6

(1 MPa = 145 psi)

When you have slipping of crystalline planes in a real crystal (with dislocation), a lower amount of chemical bonds will break compared to a perfect crystal (without dislocations) → **lower yield stress**

Technological implications?



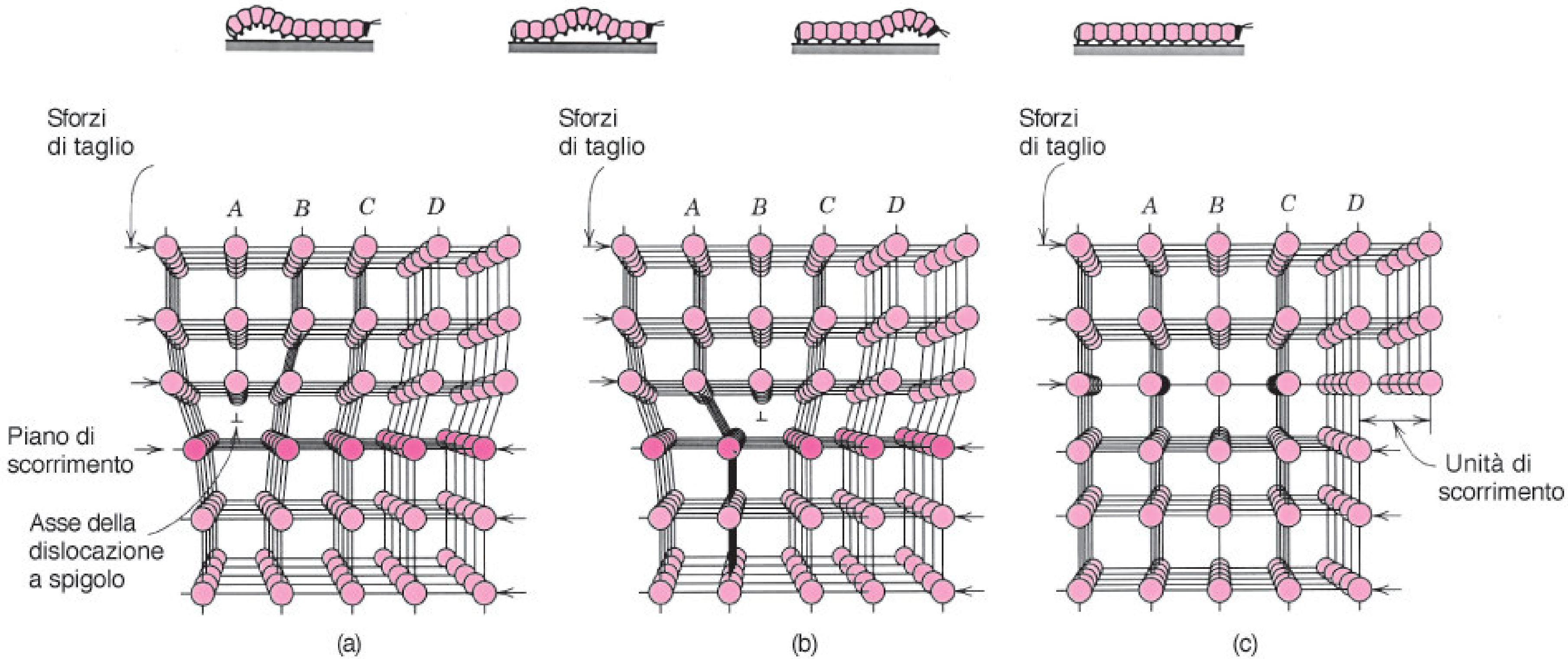
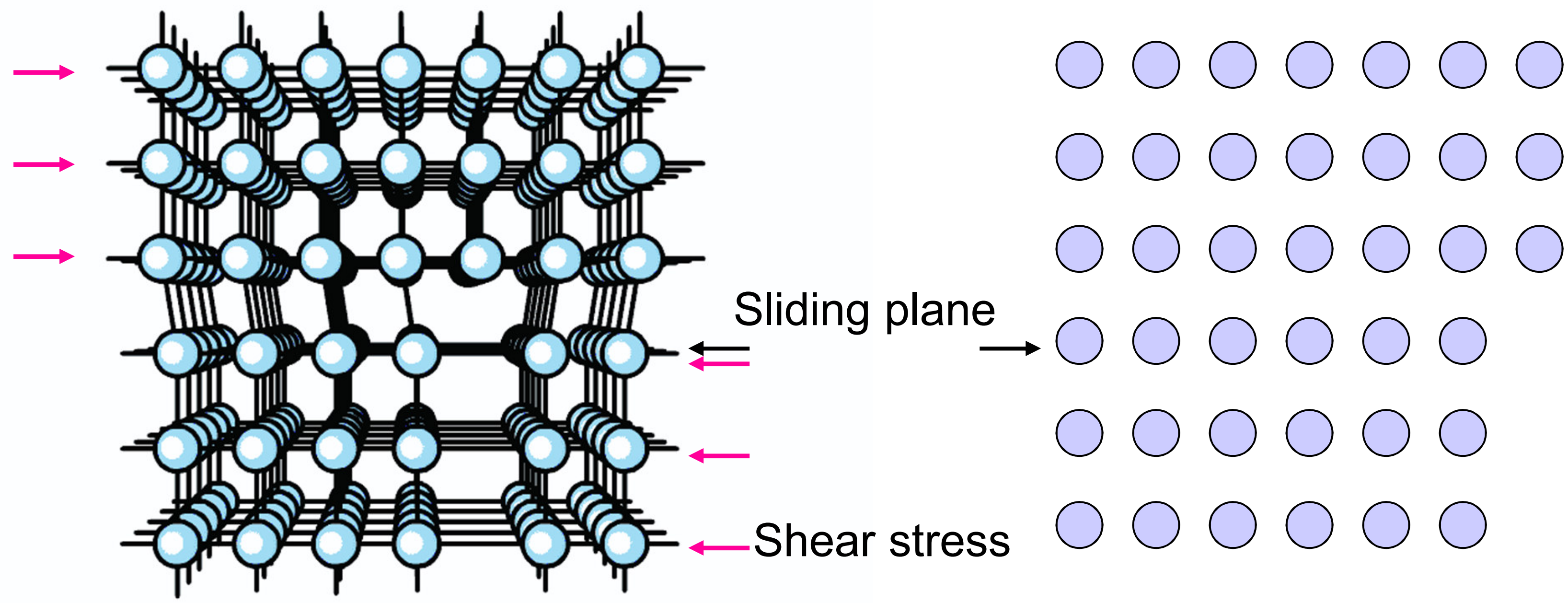


FIGURA 7.1 Ridisposizione degli atomi causata dal moto di una dislocazione a spigolo sollecitata da uno sforzo di taglio. (*a*) Il semipiano aggiuntivo è indicato con *A*. (*b*) La dislocazione si sposta a destra di una distanza interatomica non appena *A* si unisce alla parte inferiore del piano *B*; nel processo, la parte superiore di *B* diventa il semipiano aggiuntivo. (*c*) Quando il semipiano aggiuntivo raggiunge la superficie del cristallo si forma un gradino. (Da A.G. Guy, *Essenzial of Materials Science*, McGraw-Hill Book Company, New York, 1976, p. 153.)

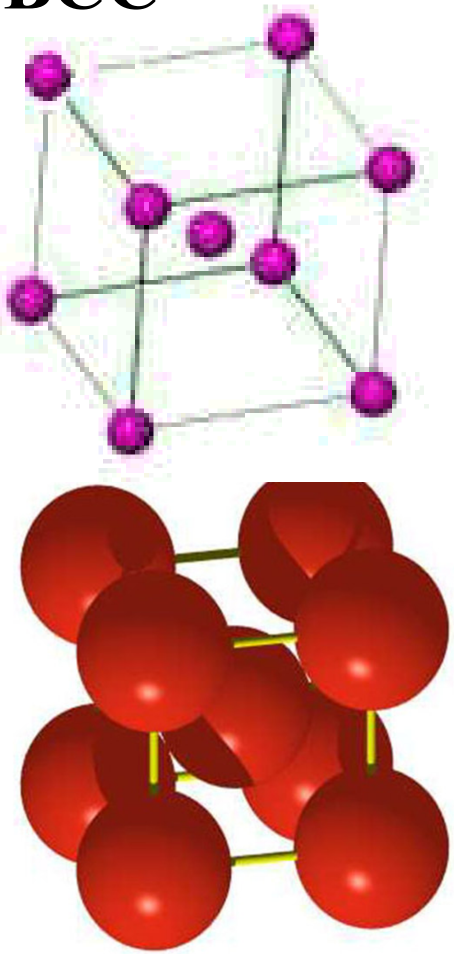
Callister
Scienza e Ingegneria dei materiali, una introduzione
EdiSES

Dislocations provide an effective way for bonds to break and re-form in another place, when load is applied.



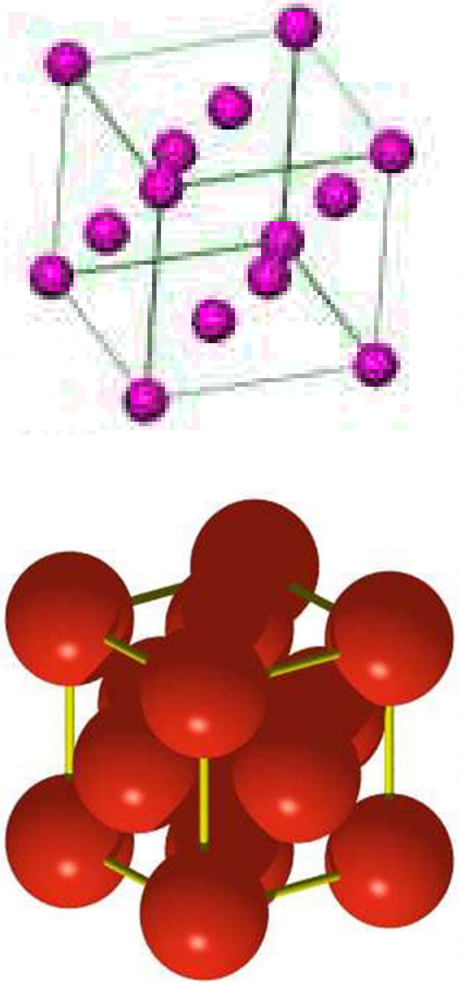
Dislocation motion is favoured on high density sliding planes

BCC



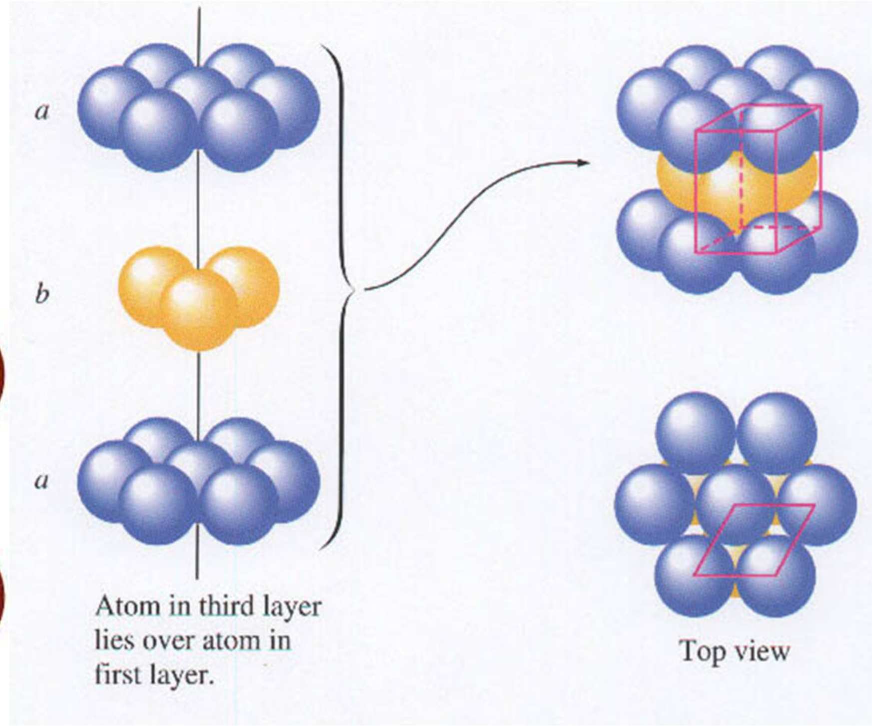
Es: Fe, W, Cr

FCC

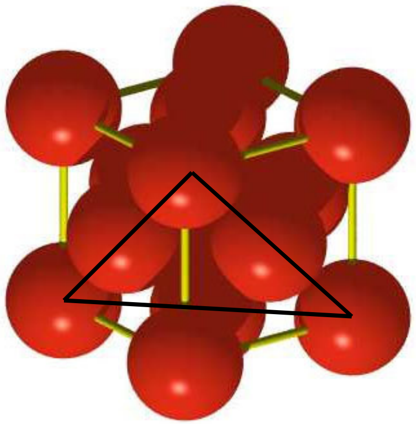


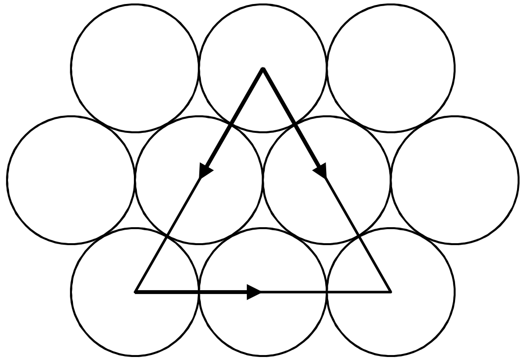
Es: Cu, Al, Ag, Au

Hexagonal compact



Es: Mg, Ti, Zn

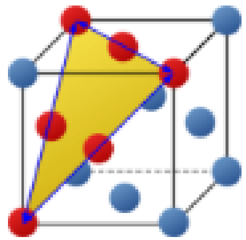
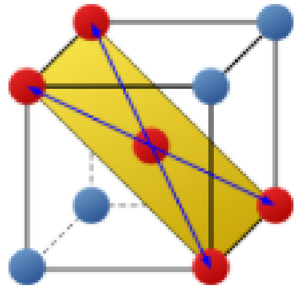
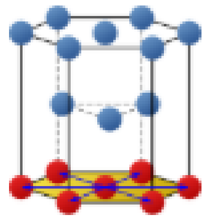




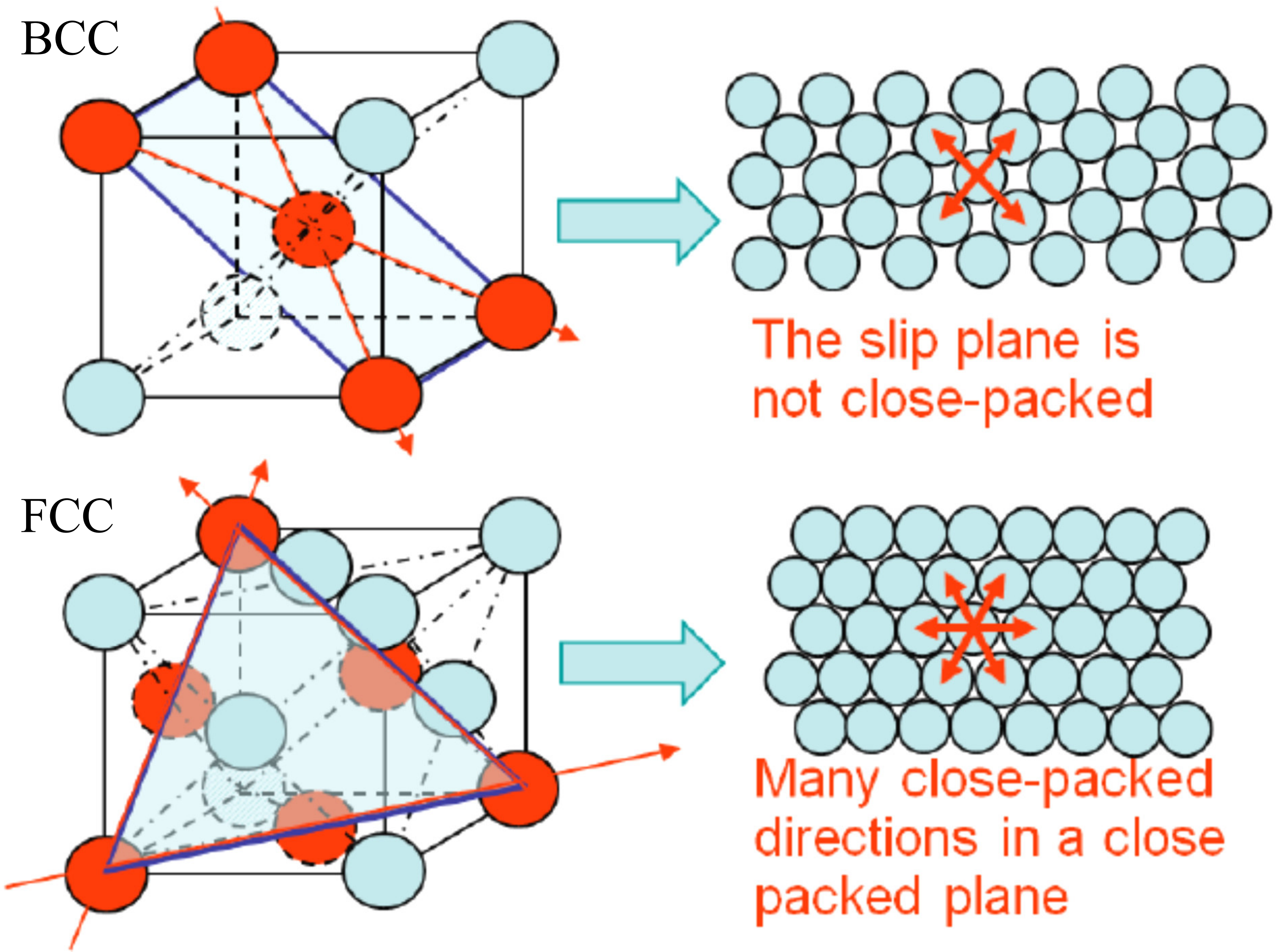
plane (111) for CFC,
with three sliding
directions <110>

- BCC e FCC have at least 12 sliding planes
- HCP has a maximum of 6 sliding planes
- FCC metals (Cu, Al, Ni, Ag, Au) and BCC (Fe) are ductile, with large plastic deformation
- HCP metals (Ti, Zn) are more brittle

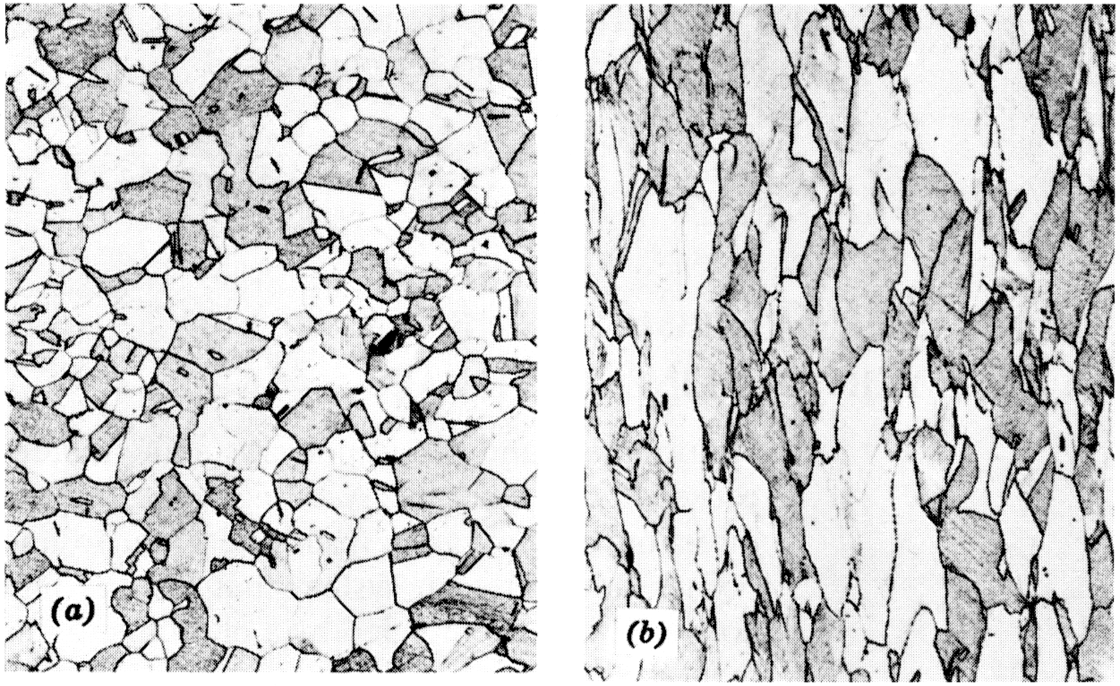
- Number of preferred slip systems
  - https://www.tec-science.com/material-science/ductility-of-metals/influence-of-the-lattice-structure-on-the-ductility/
  - FCC (Cu, Al, Ni, Ag, Au) and BCC (Fe) metals have large plastic deformation
  - HCP (Ti, Zn) metals have smaller plastic deformations

	face-centered cubic (fcc)	body-centered cubic (bcc)	hexagonal closest packed (hcp)
lattice structure			
number of preferred slip planes	4	6	1
number of slip directions	3	2	3
number of preferred slip systems	12	12	3
packing density	74 % (densest packed)	68 % (loosely packed)	74 % (densest packed)
ductility	high	medium	low
metals	γ-iron (<1392°C), aluminium, β-cobalt (>417°C), lead, copper, nickel	α-iron (<911°C), chrome, molybdenum, β-titanium (>882°C), vanadium, tungsten	α-cobalt(<417°C), α-titanium (<882°C), zinc, magnesium

For good deformability, a lattice structure should have at least 5 slip systems to be activated during a deformation process, so that the workpiece can deform in any direction as desired.


BCC
The slip plane is
not close-packed
FCC
Many close-packed
directions in a close
packed plane

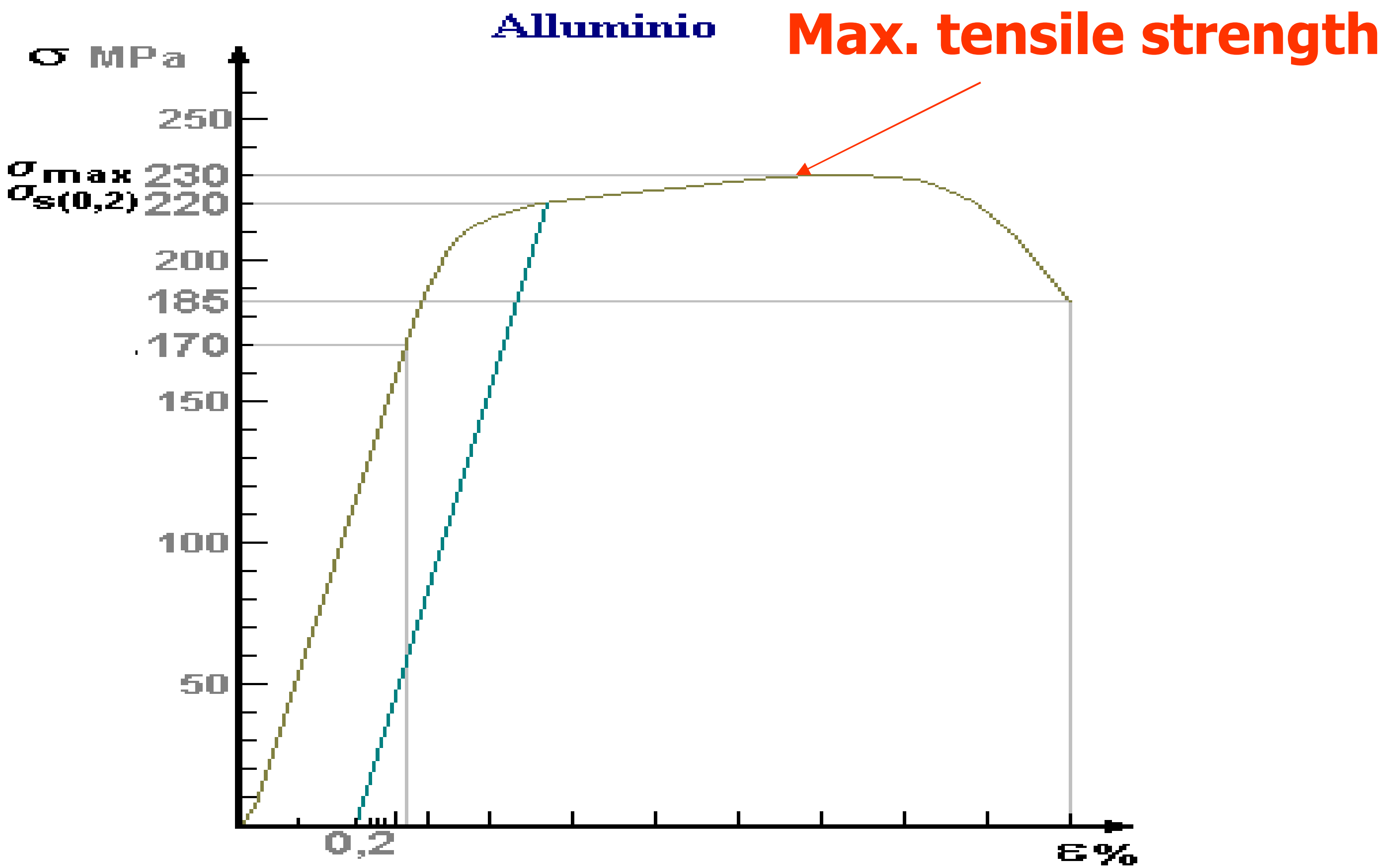
- Plastic deformation of grains in a stainless steel after lamination



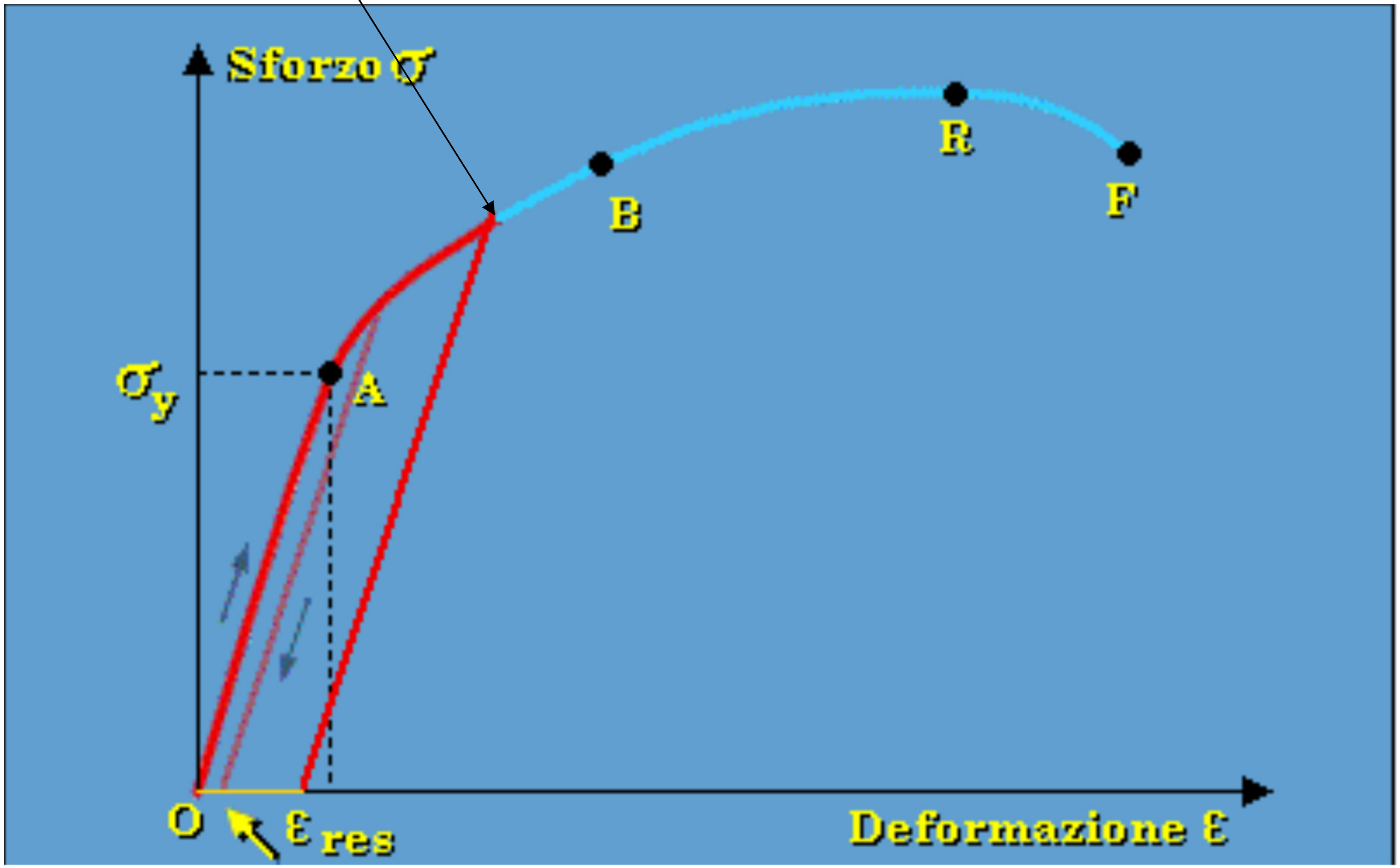
After deformation

from Callister

Stress-strain curve for Al



If we stop loading here a residual plastic deformation (ε_{res}) is still present



Plastic (residual, permanent) deformation

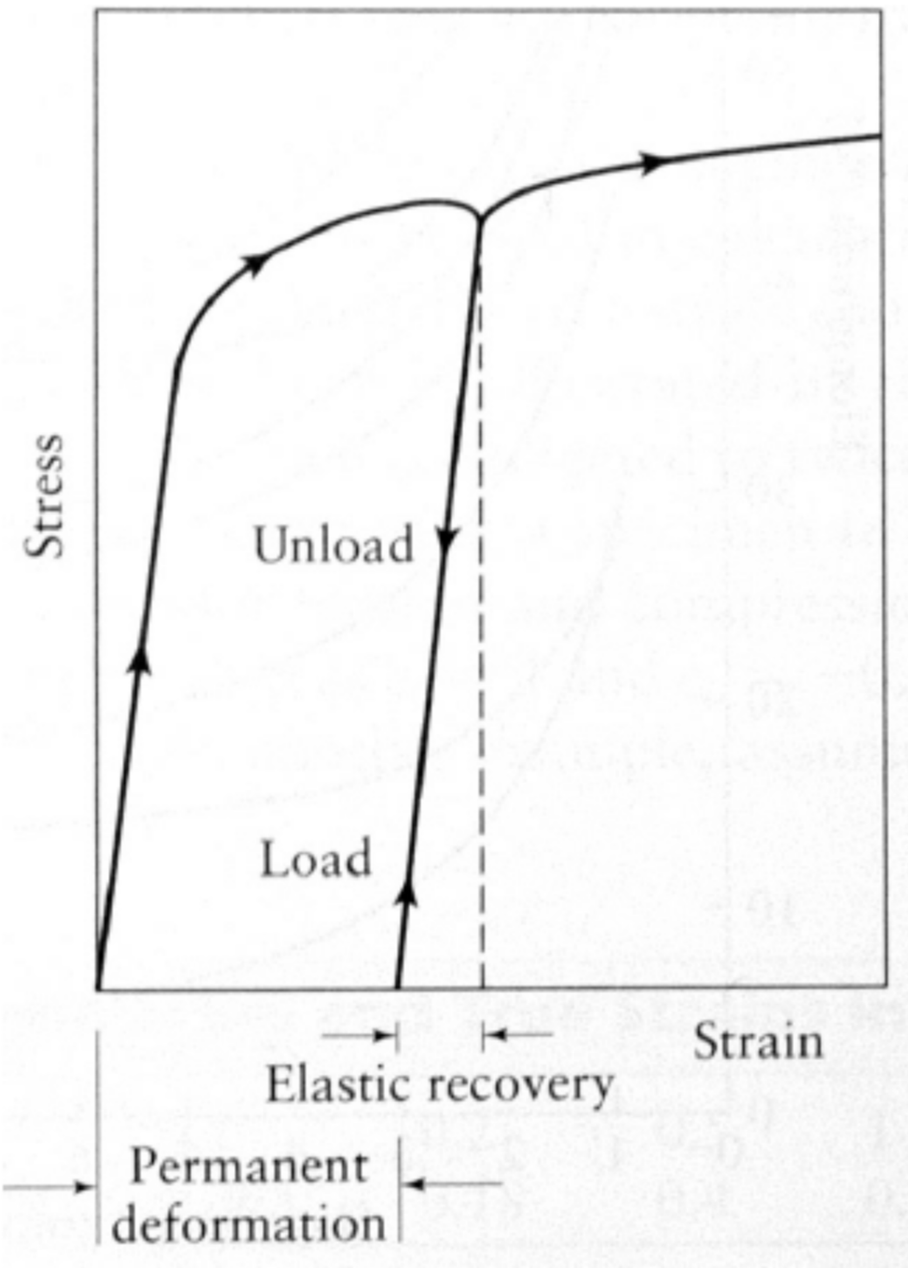
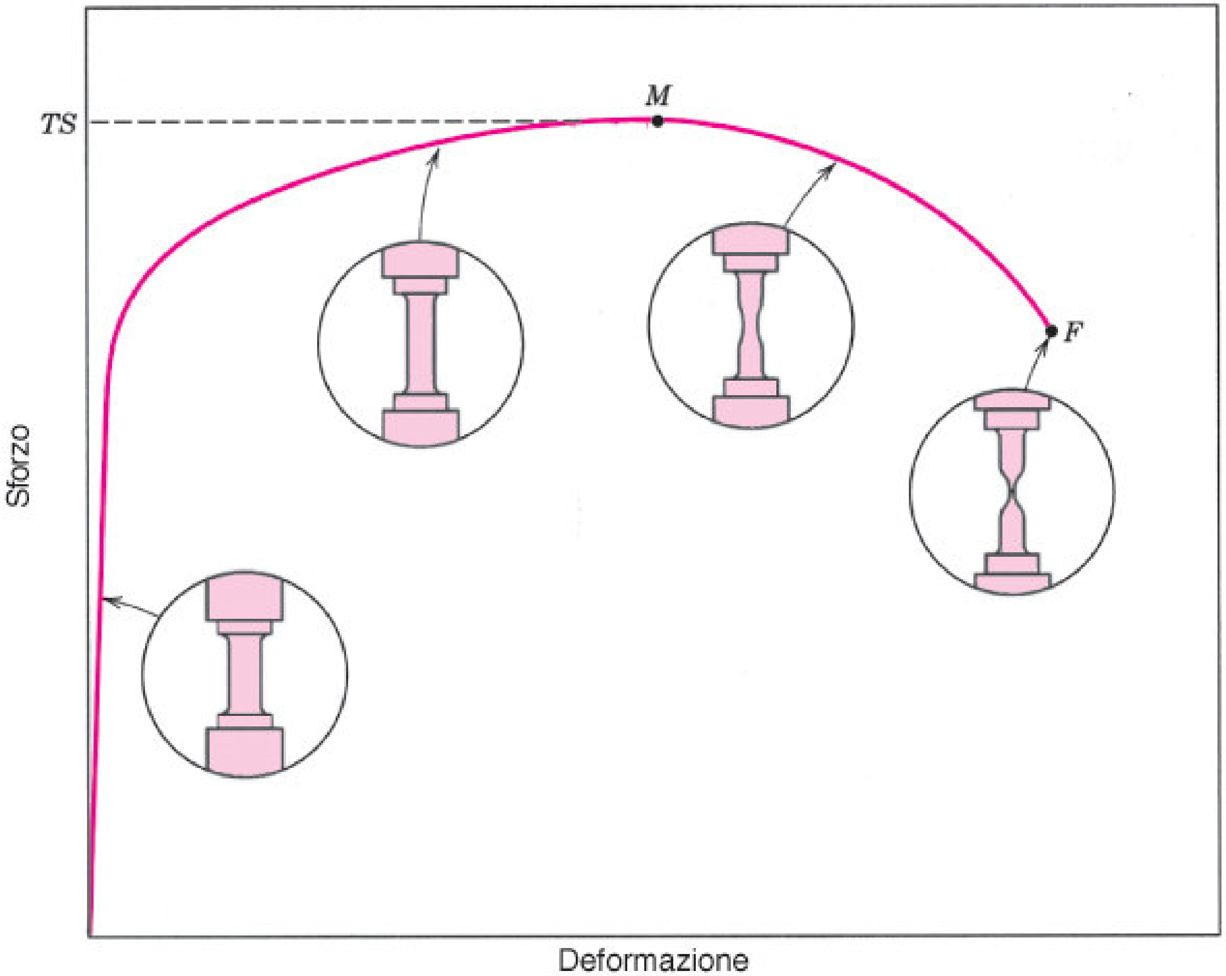


FIGURA 6.11 Tipica curva sforzo–deformazione fino a frattura (punto *F*). Il carico di rottura è indicato dalla lettera *M*. Nei riquadri circolari sono rappresentate le variazioni della geometria del provino in punti differenti della curva.



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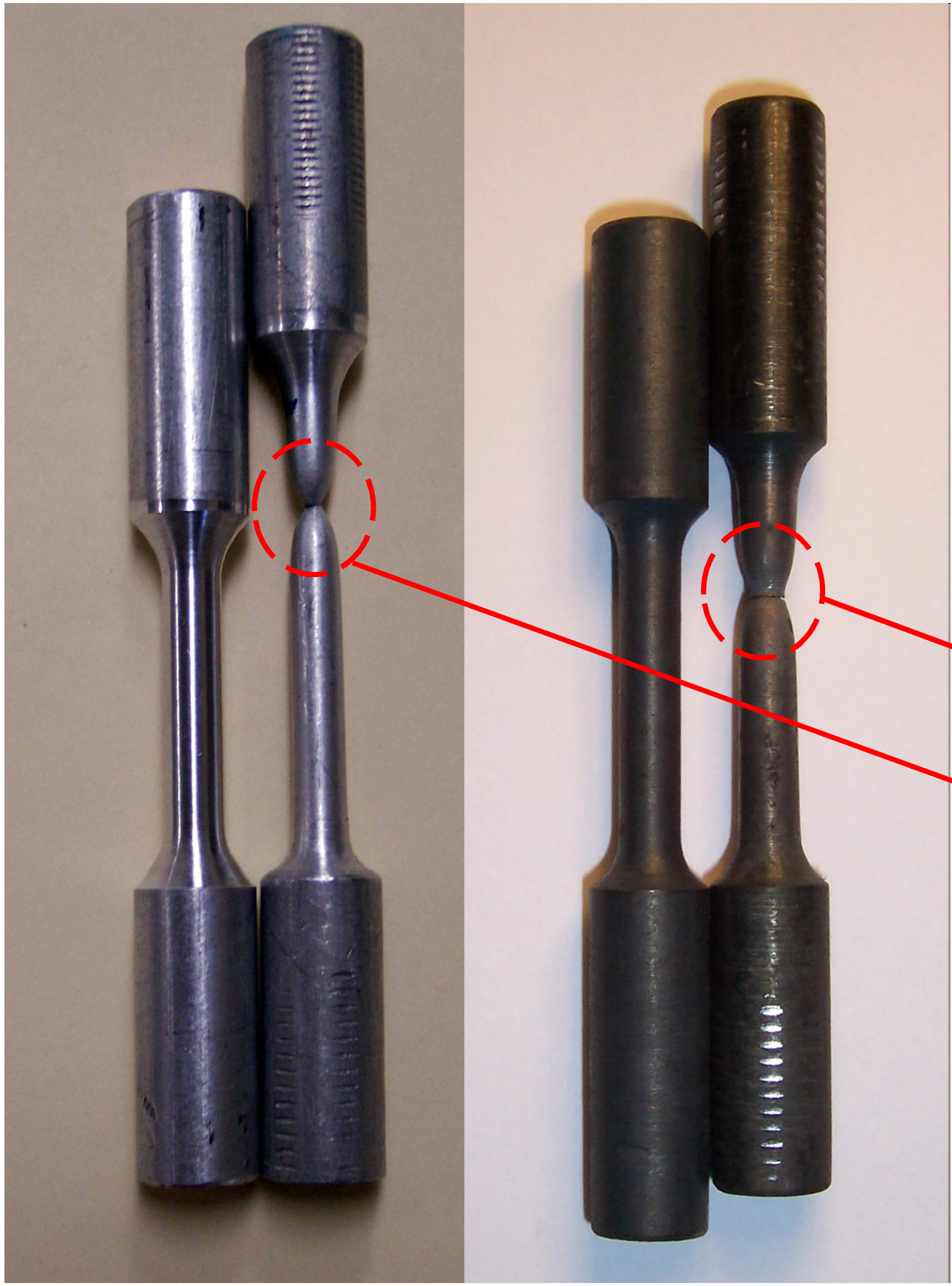
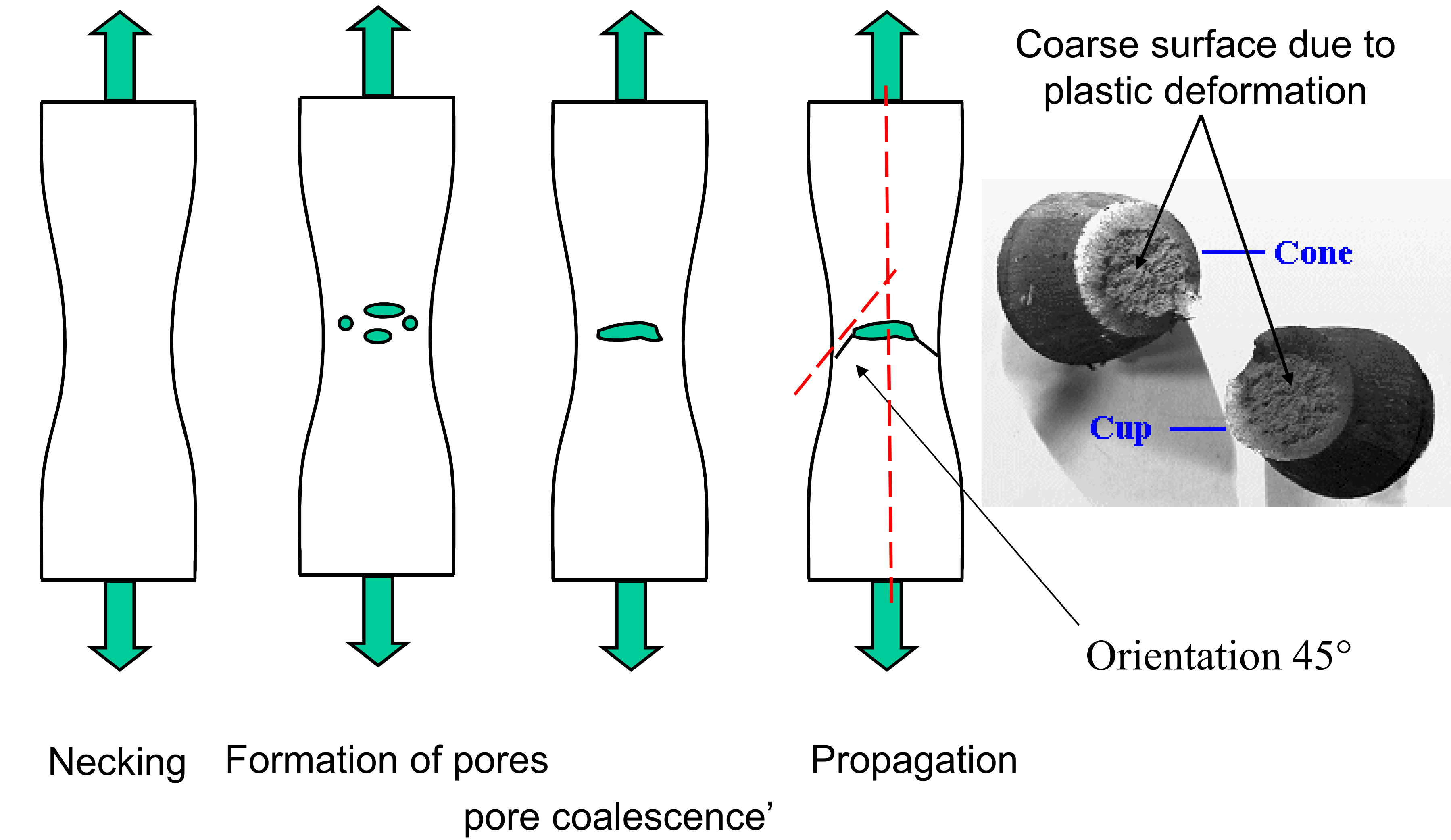


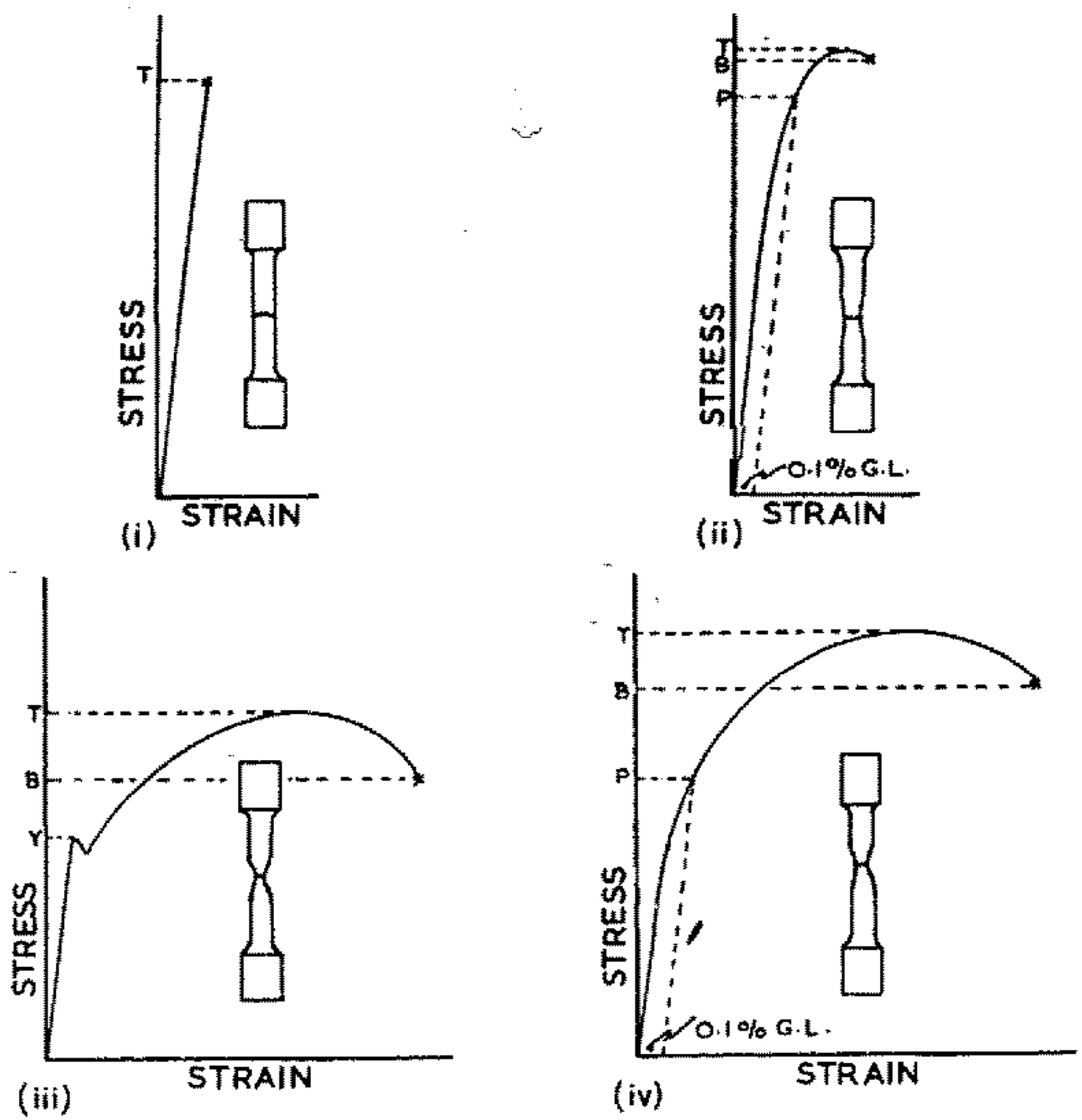
Metal specimens of steel (left) and bronze (right) before and after the test

NECKING:
reduction of the resistant section (lateral contraction) due to a rearrangements of the material structure

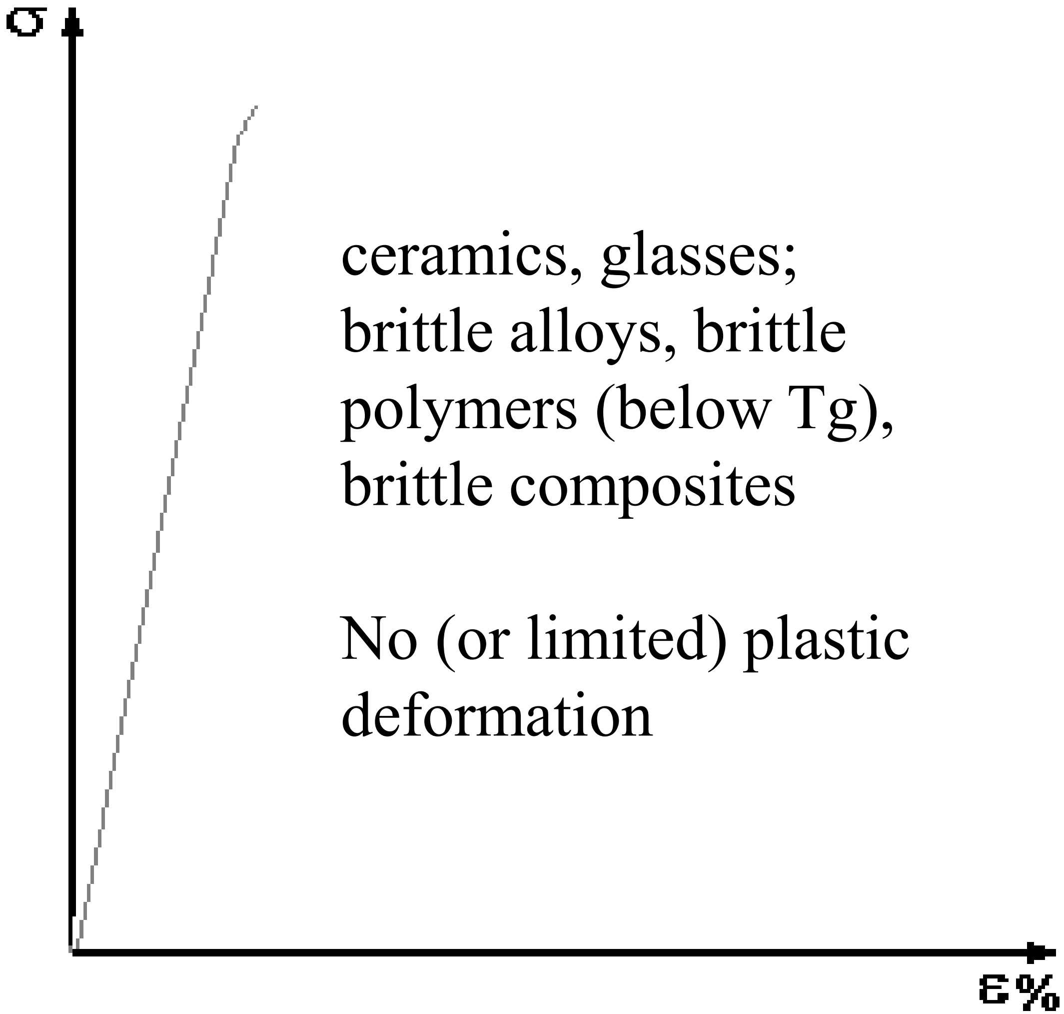
Ductile fracture



Tensile strength behaviour for different materials



Stress-strain curves for brittle materials

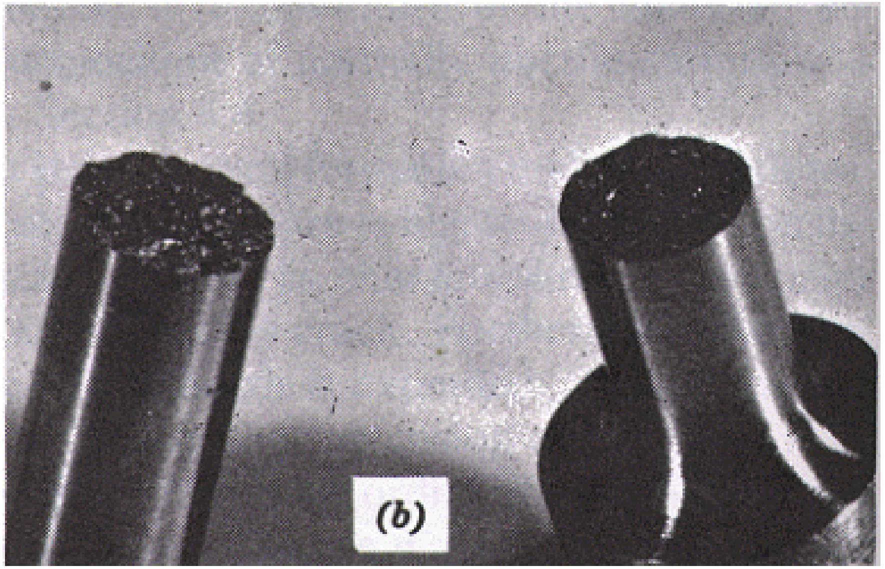


ceramics, glasses;
brittle alloys, brittle polymers (below Tg), brittle composites

No (or limited) plastic deformation

Brittle fracture

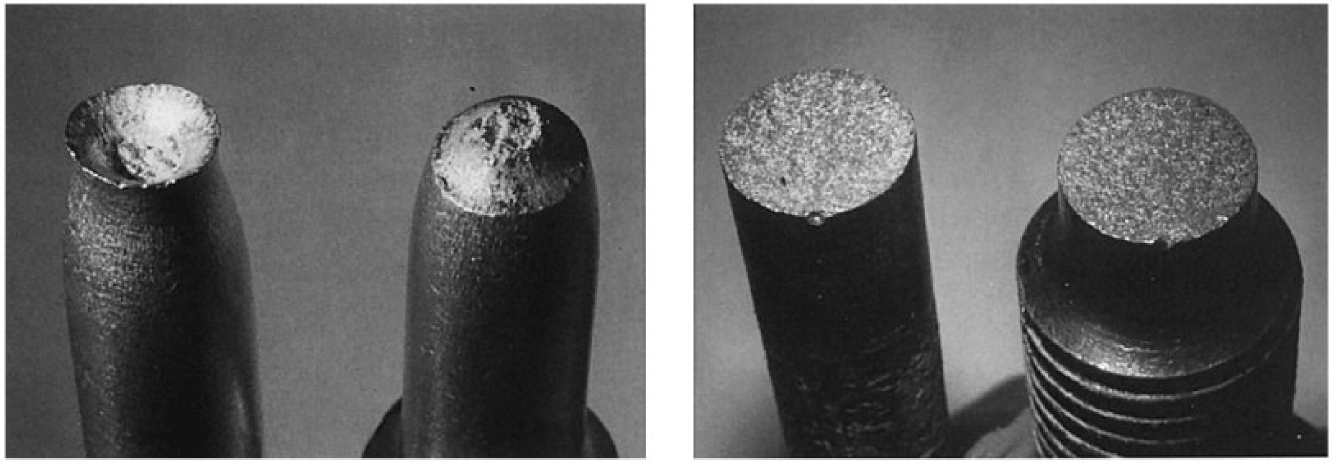
- Without plastic deformation
  - Flat fracture surfaces



  - Along crystalline planes for mono-crystalline materials (cleavage)
  - Intergranular fracture for poly-crystalline materials

Ductile and brittle fracture surfaces

- Ductile fracture: plastic deformation at the crack edge, slow crack propagation
- Brittle fracture: no plastic deformation at the crack edge, quick crack propagation (catastrophic failure)



(a) (b)

FIGURA 8.3 (*a*) Frattura coppa e cono nell'alluminio. (*b*) Frattura fragile in un acciaio dolce.

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Tensile strength (σ) and % ε deformation at fracture

Material	σ (MPa)	ε (%)
steel	400-2500	40-12
Al alloys	300-770	8-16
Al_2O_3 sint.	200-340	negligible
SiO_2 glass	110	negligible
ZrO_2 sint.	80	negligible
Polyesthers	40-80	1-2
Nylon	70-160	50-200
Epoxy	30-120	3-6

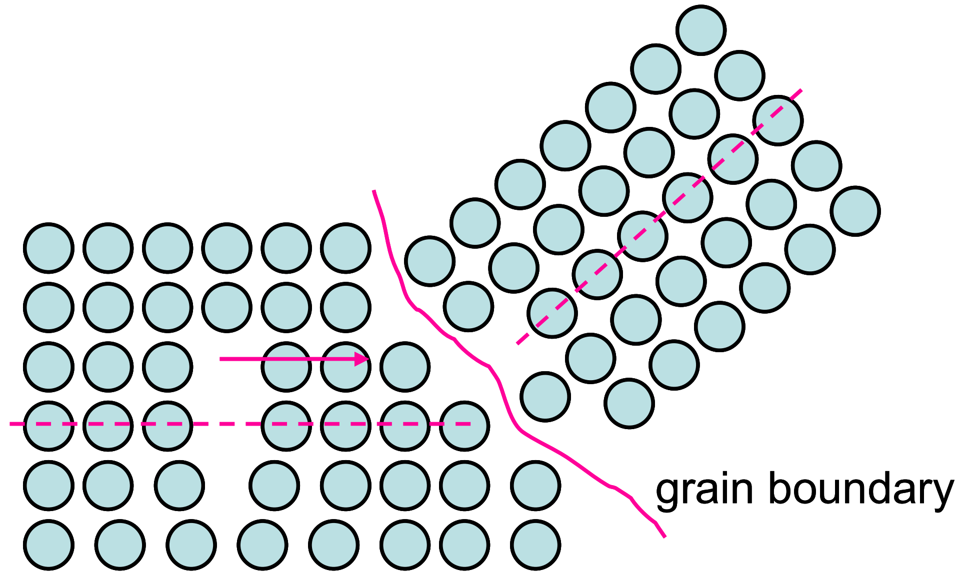
How to modify plastic properties of materials

- Make dislocation motion more difficult = increase materials mechanical strength (and hardness)

- Grain size
- Solid solutions
- Precipitates and second phases
- Work hardening
- Perfect crystals/monocrystals

Grain size reduction

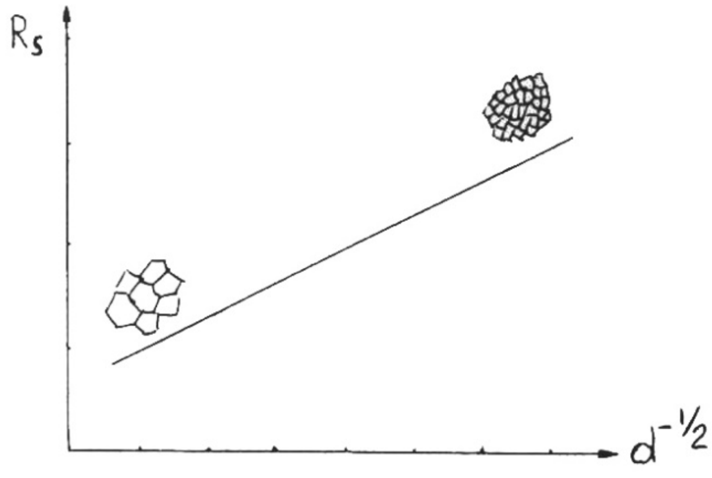
$$\sigma_y = \sigma_0 + k_y d^{-1/2}$$

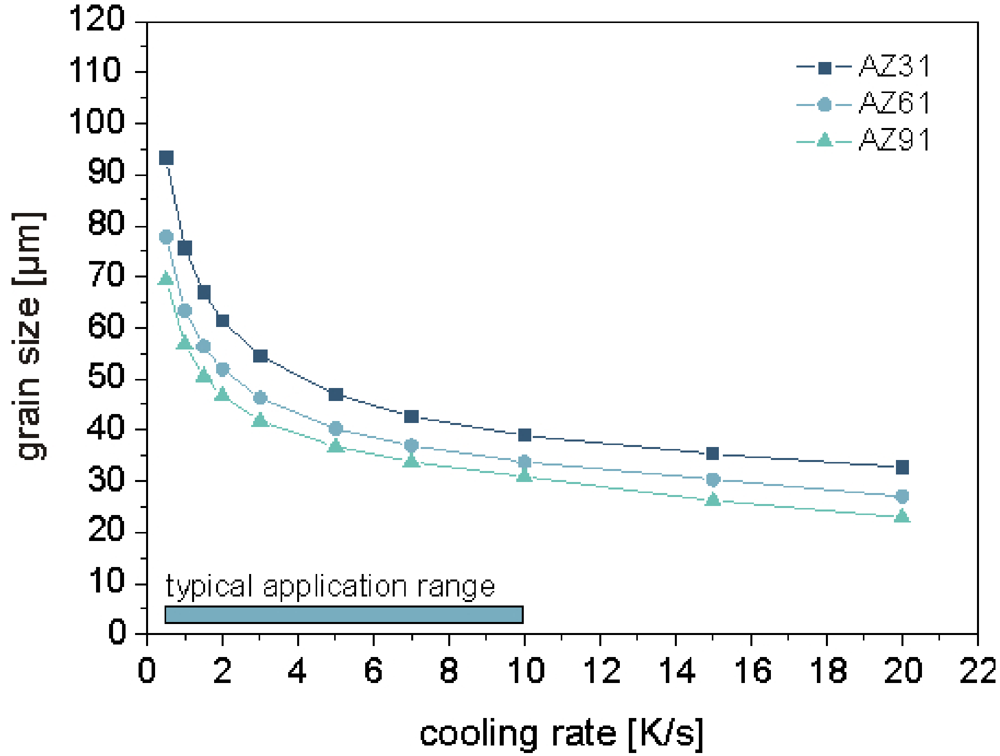


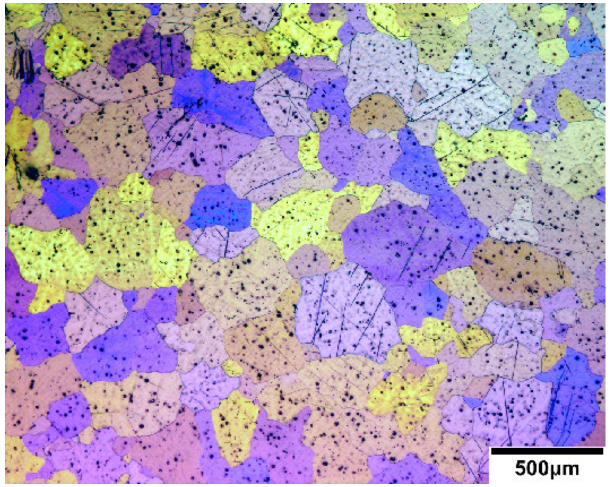
Hall-Petch law

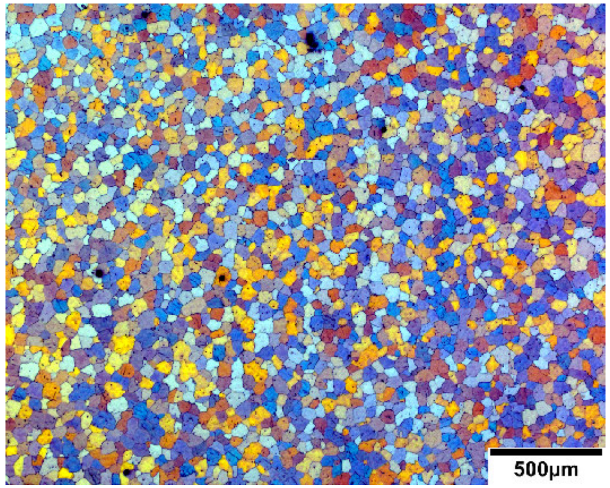
d = grain diameter (avg)
σ_y = yield strength
Materials related costant

Grain size can be modified by thermal treatment









Need for nucleating agents
(fine ceramic powders in the melt)

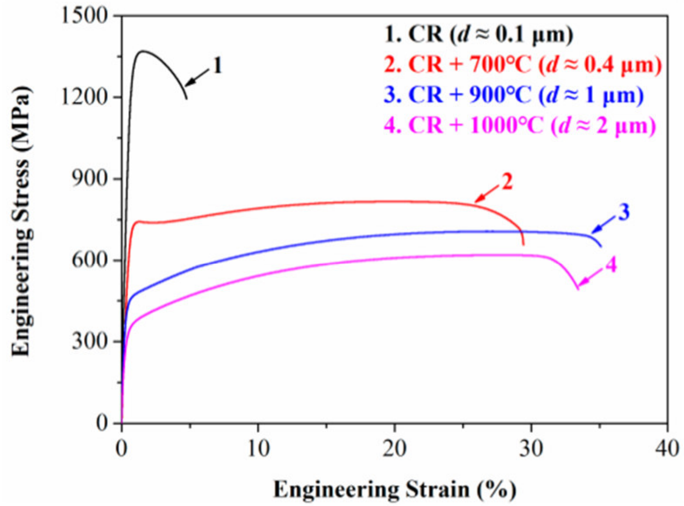


Figure 4. Engineering stress-strain curves of samples with different grain size

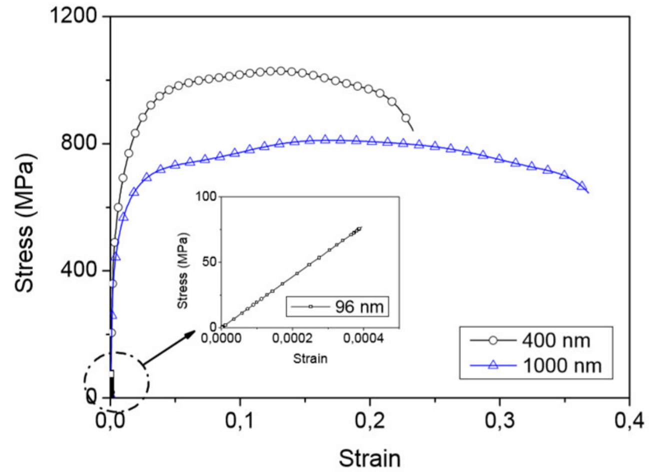


Fig. 5. Tensile stress-strain curves for the different grain sizes studied

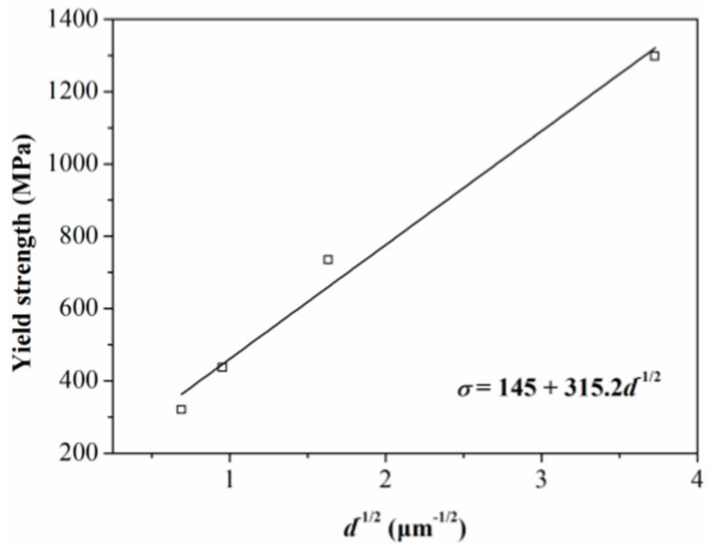
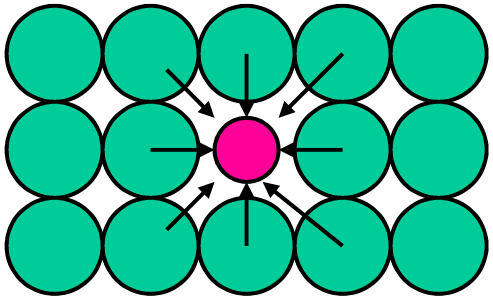


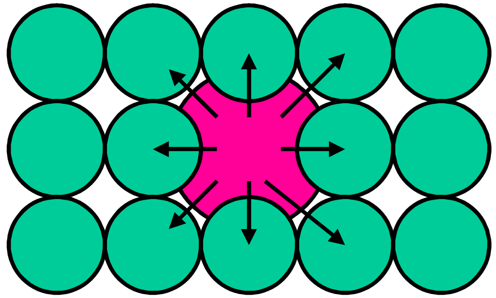
Figure 8. Yield strength vs. the inverse square root of grain size.

Solid solutions

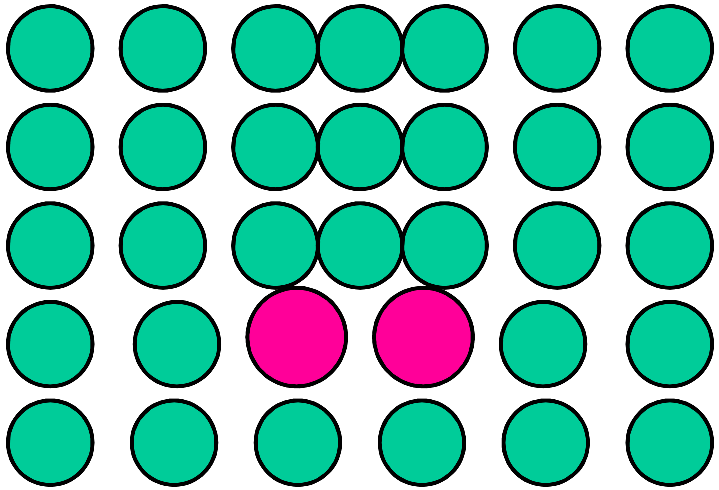
- Metallic solid solutions (alloys) have superior mechanical strength than pure metals



Tensile stresses



Compression stresses



Dislocations motion slowed

Role of alloying elements on mechanical properties

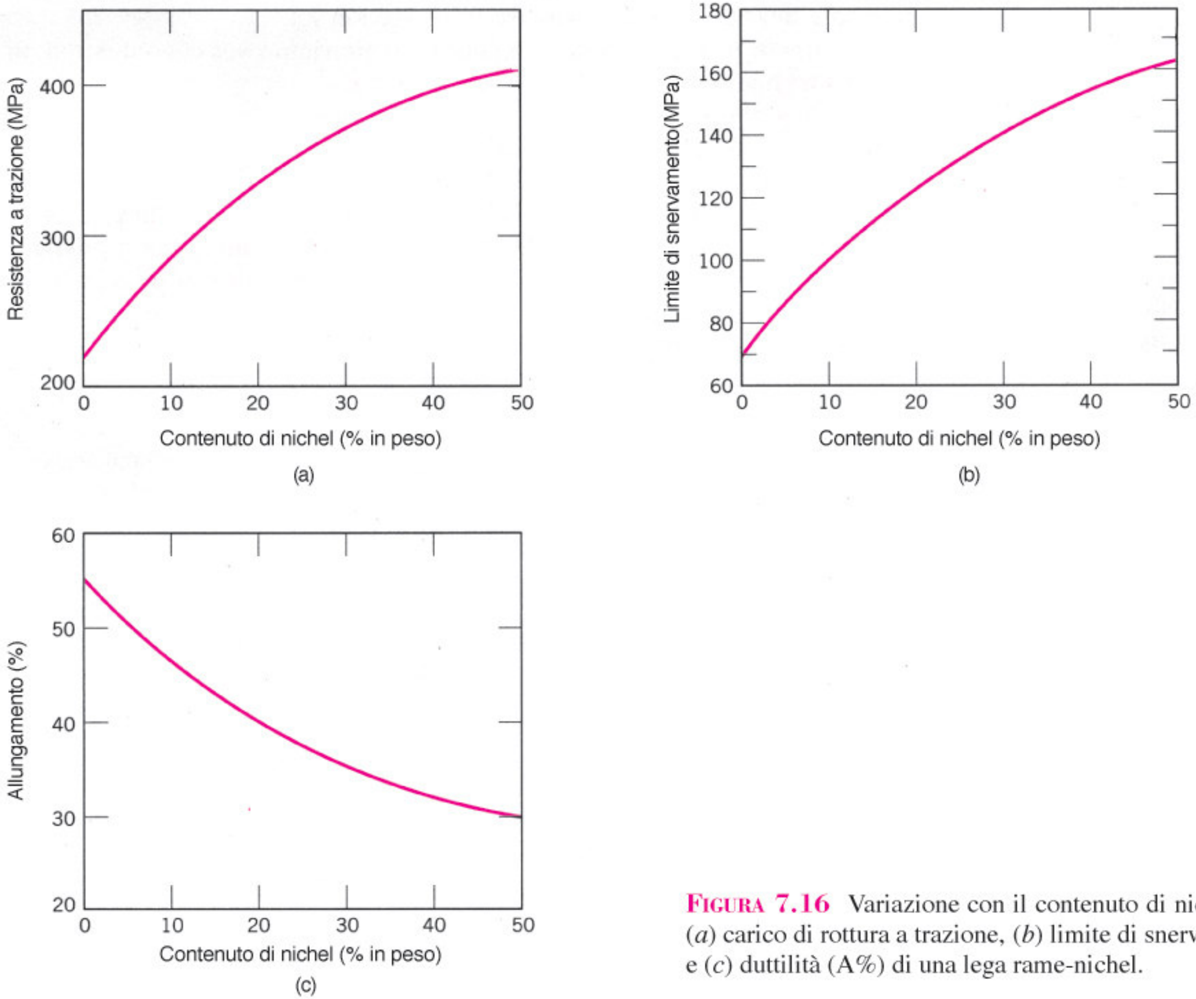


FIGURA 7.16 Variazione con il contenuto di nichel del (*a*) carico di rottura a trazione, (*b*) limite di snervamento e (*c*) duttilità (A%) di una lega rame-nichel.

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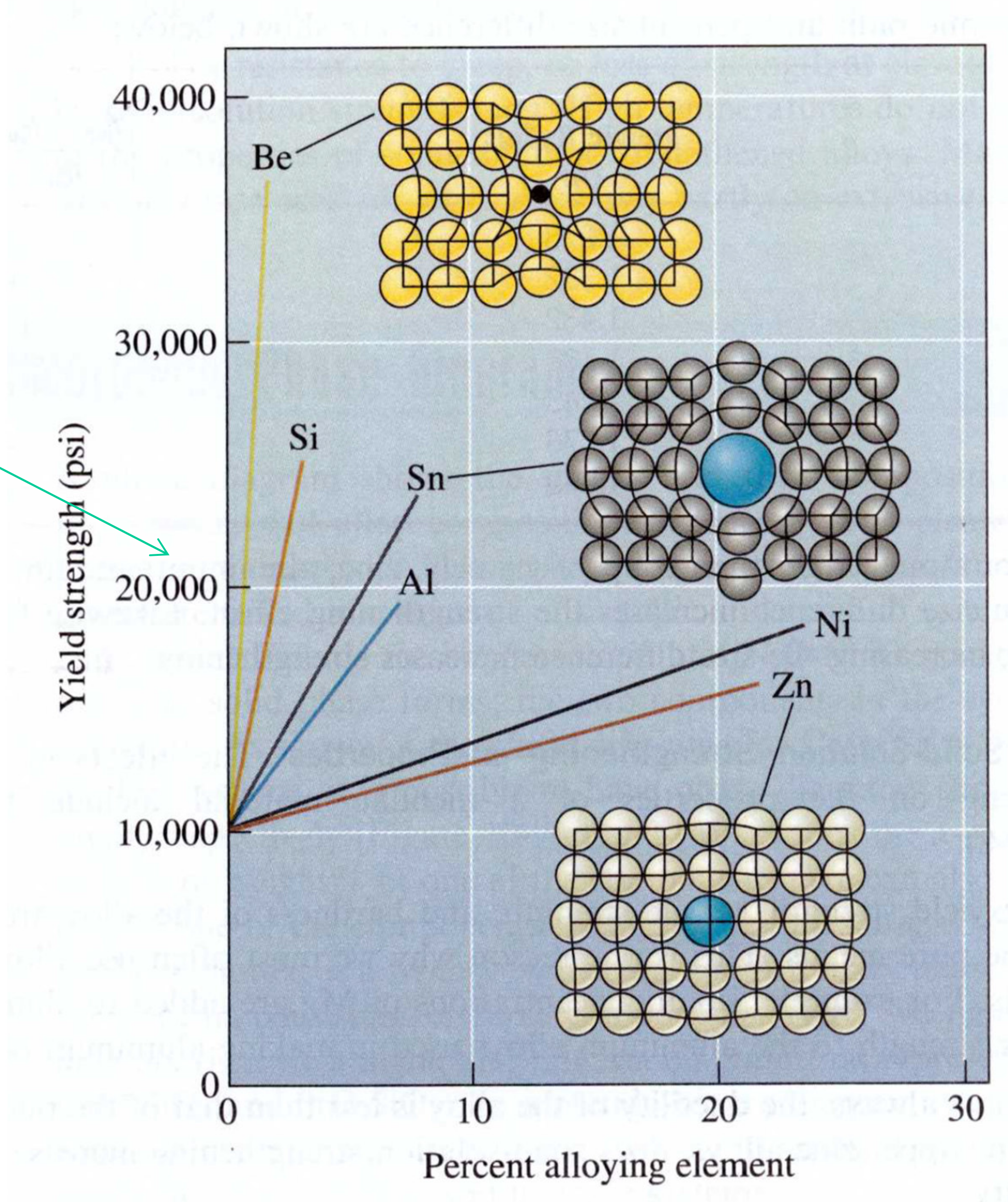
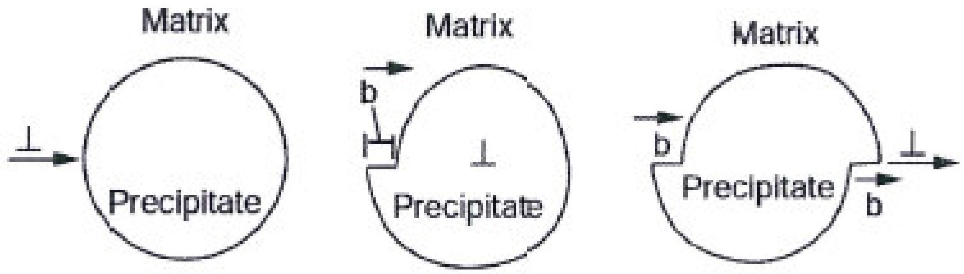
1 MPa = 145 psi; 20000 psi = 137.9 MPa

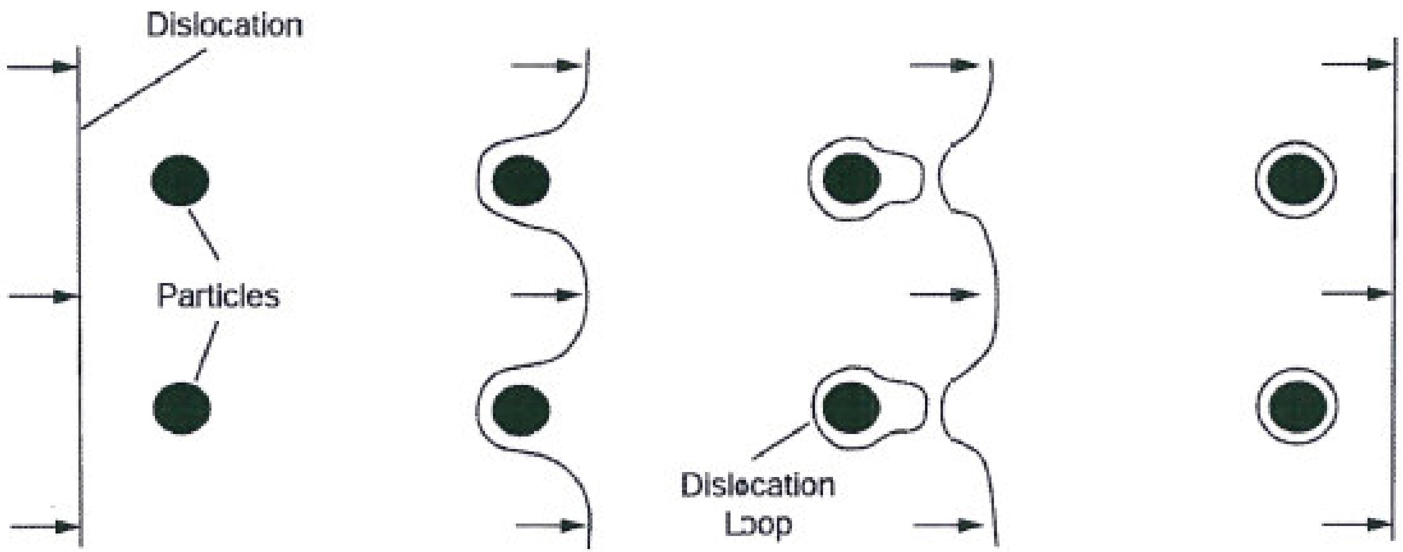


The efficacy in increasing the yield strength is related to the solute/solvent size ratio

Second phases

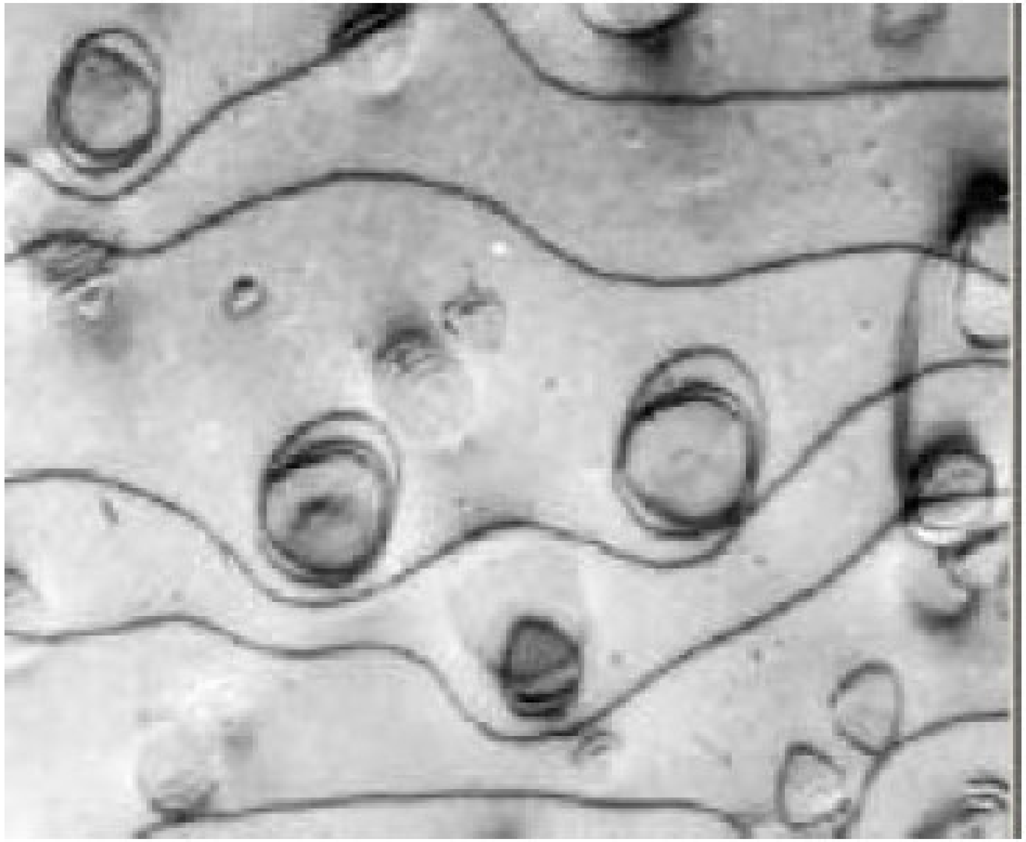


Dislocation Shearing Through Particle



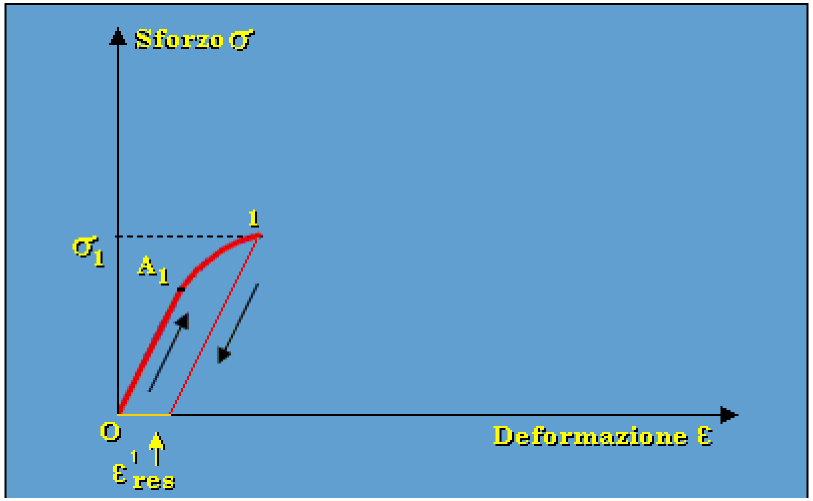
Dislocation Looping Around Particles

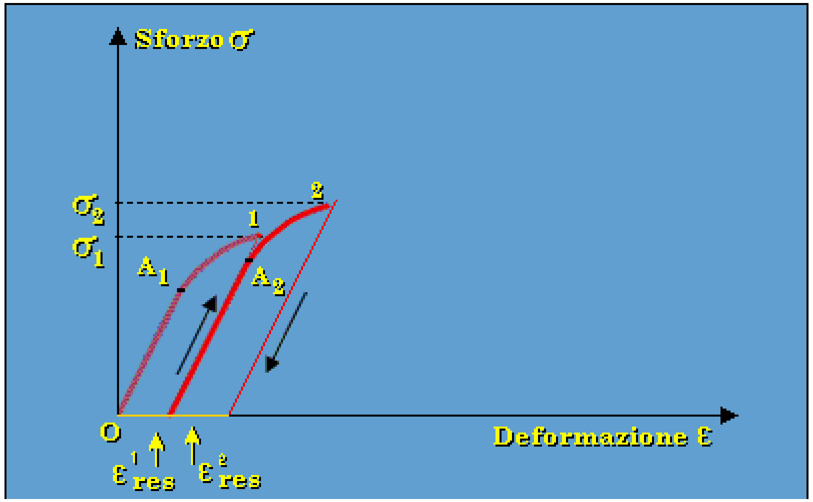
Orowan mechanism → ring dislocation around rigid inclusions



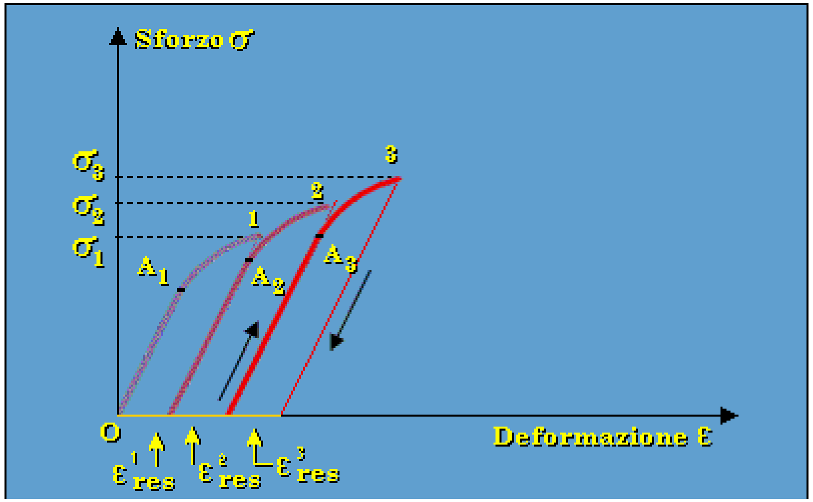
The efficacy of the mechanism is related to the number of particles rather than the total mass of particles.

Work hardening





Dislocation density: little-deformed metals: 10^6 cm/cm^3 (10,000 m/cm^3); highly-deformed metals: 10^{12} cm/cm^3 (10,000,000 km/cm^3)

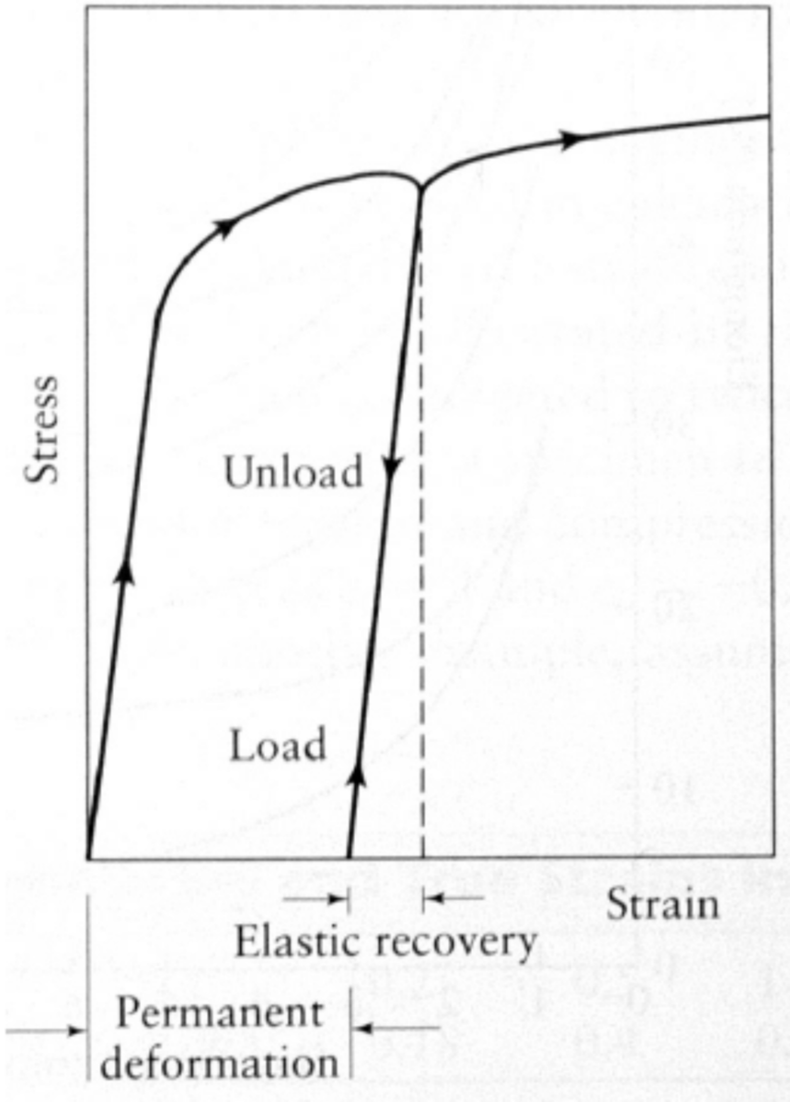


The material undergoes **<u>successive loading-unloading cylces ("cold working")</u>** → formation of new dislocations → accumulation of dislocations → barrier to further deformation → strengthening of the material → increase of **σ_y (and hardness)**.

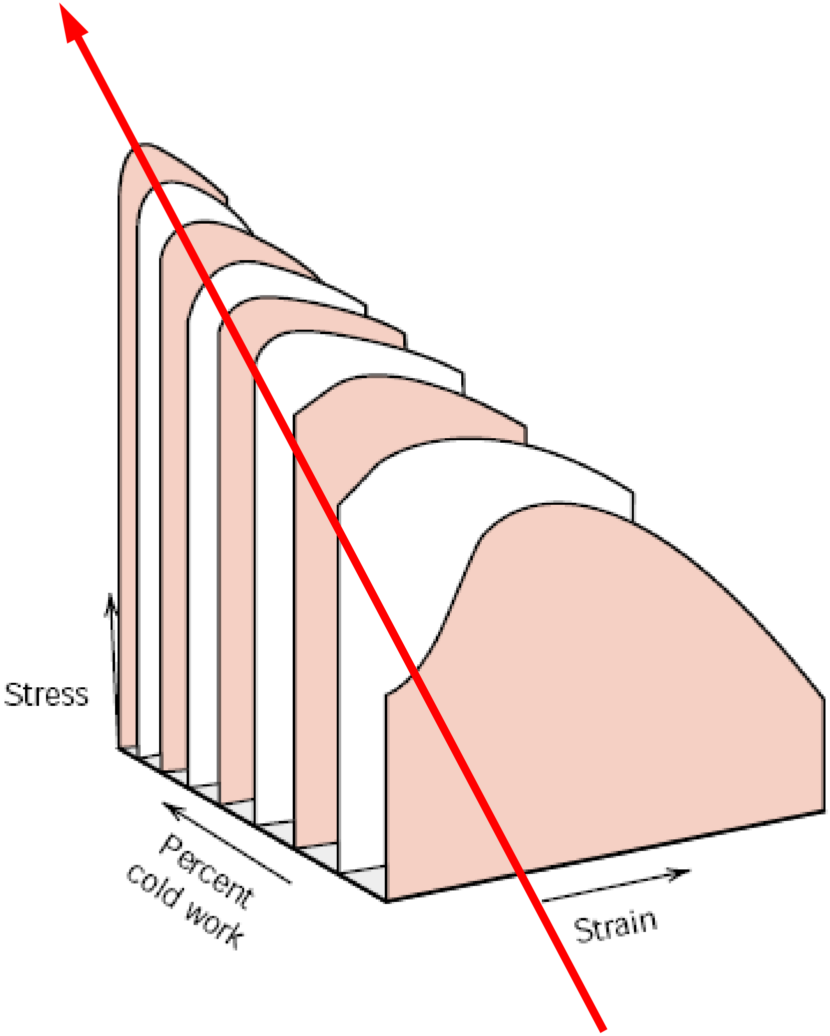
Ramberg-Osgood law
(plastic field):

$\sigma = H \times \varepsilon^n$

H and n = constant
n is called work-hardening exponent



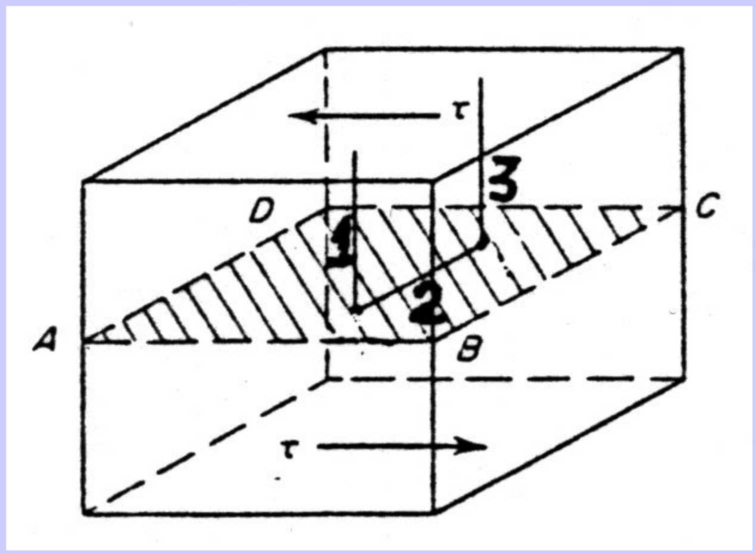
COLD WORKING OF STEELS



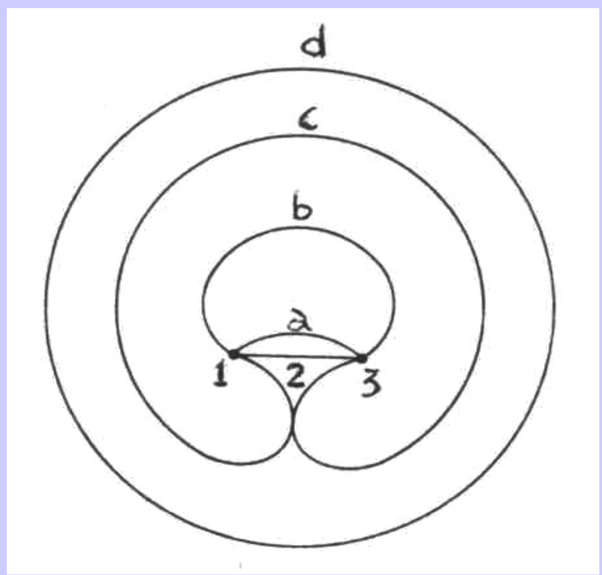
Increase of σ_y

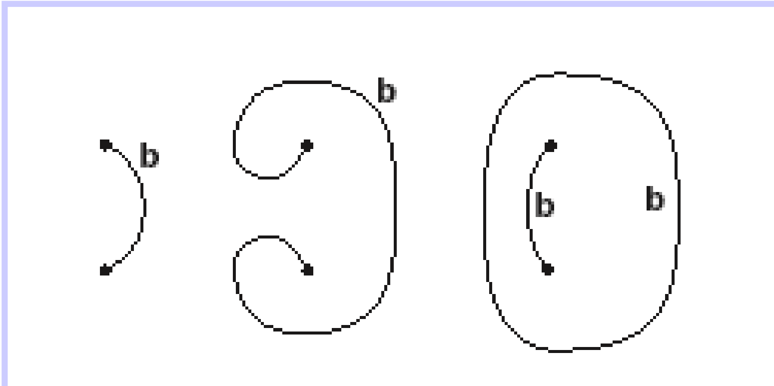
FIGURE 8.20 The influence of cold work on the stress–strain behavior for a low-carbon steel. (From *Metals Handbook: Properties and Selection: Irons and Steels,* Vol. 1, 9th edition, B. Bardes, Editor, American Society for Metals, 1978, p. 221.)

- REVERSIBLE PROCESS:
- COLD WORKING/ANNEALING

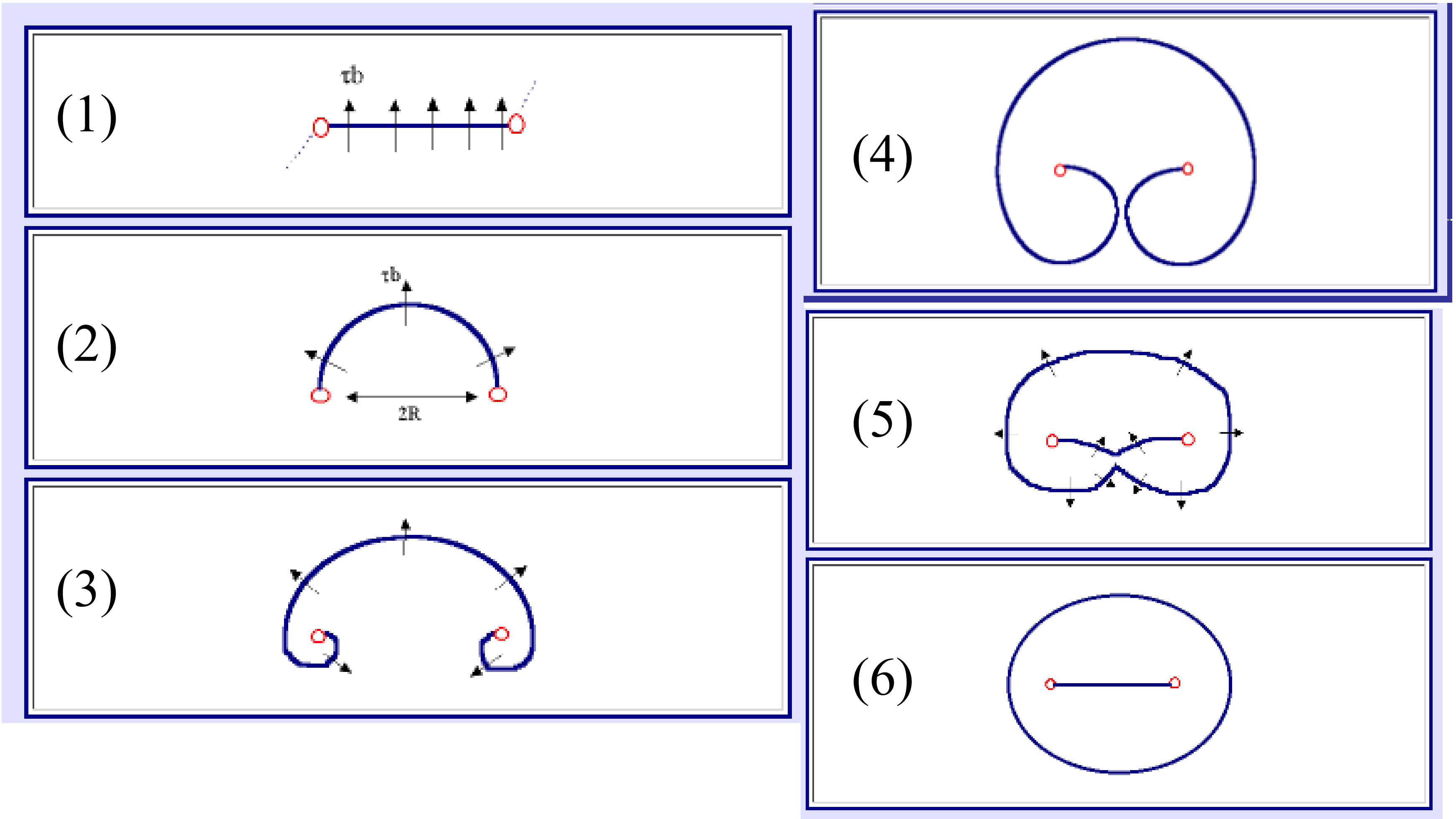


Mechanism of multiplication of dislocations («forest»)



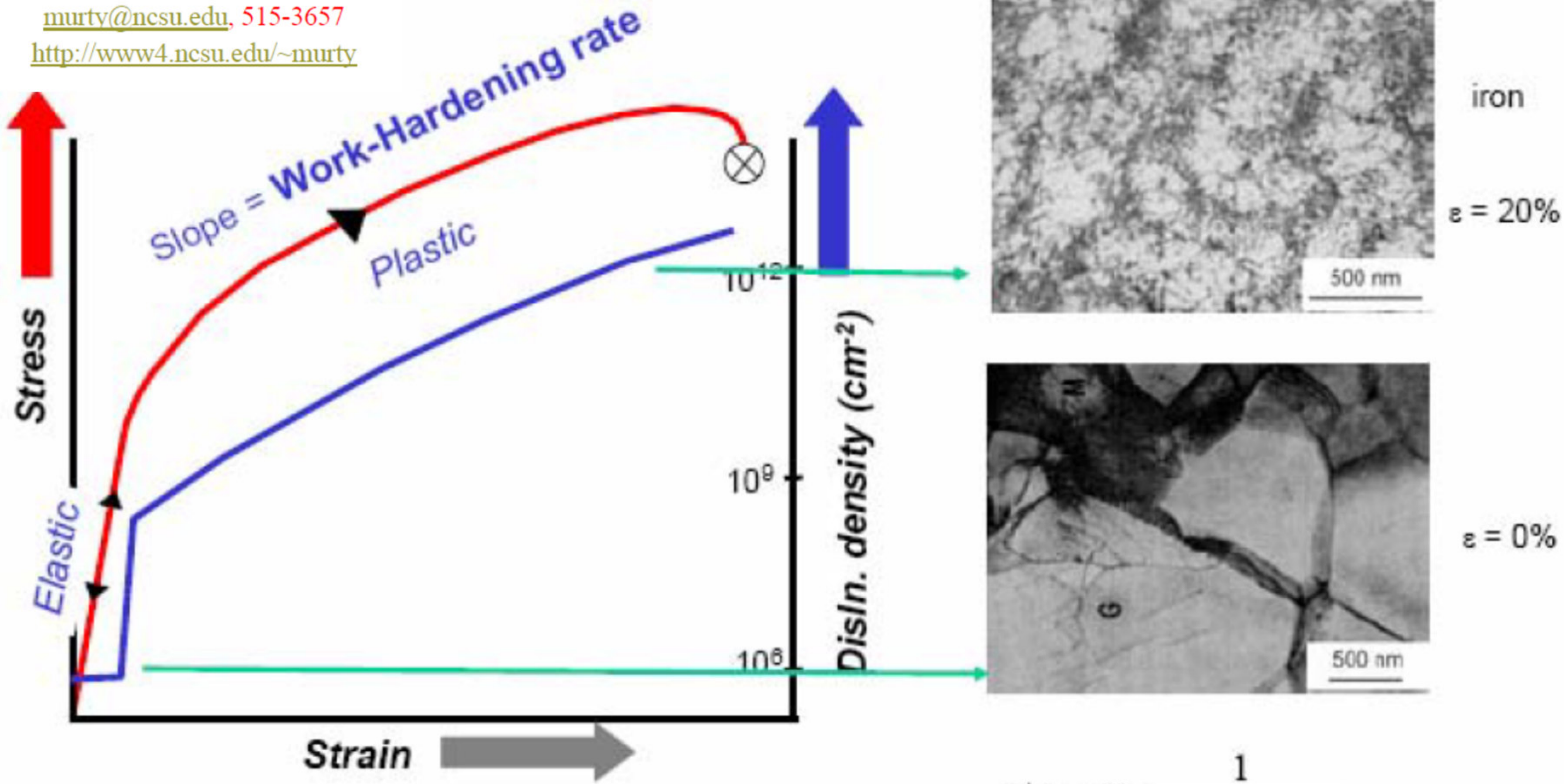


Frank-Read mechanism


(1)
τb
(2)
τb
2R
(3)
(4)
(5)
(6)

Work-hardening on metals and alloys

Prof. K. L. Murty
North Carolina State University
murty@ncsu.edu, 515-3657
http://www4.ncsu.edu/~murty



Most dislocations are not mobile –
they are just obstacles to the few that are.

$$\text{stress} \propto \frac{1}{\text{disln. spacing}} = \sqrt{(\text{disln. density})}$$

https://www.tec-science.com/material-science/ductility-of-metals/fundamentals-of-deformation/

1. Visualize different mechanism of hardening in stress-strain curves:

https://www.tec-science.com/material-science/ductility-of-metals/deformation-process-in-real-crystal-structures/

1. Visualize difference between Elastic deformation and plastic deformation:

https://www.tec-science.com/material-science/ductility-of-metals/deformation-process-in-real-crystal-structures/